

Designing Carbon Policy for a Biochar Supply Chain: Market Clearing, Endogenous Carbon Prices, and Technology Choice in Wisconsin

Abstract

Biochar is among the most scalable engineered carbon dioxide removal (CDR) pathways, yet realizing its potential requires supply chains whose economics are dominated by carbon accounting rather than energy sales. We develop a multi-product biochar supply chain network design model for Wisconsin (72 county nodes, 13 feedstocks, dehydration plus slow pyrolysis at 300 and 500 °C, price-responsive biochar demand) in the market-clearing framework of Sampat et al. [18], in which the duals of mass balances are spatial equilibrium prices. We embed three carbon policy architectures: (A) a baseline-and-credit mechanism whose credits scale with throughput; (B) a cap-and-trade scheme in which the carbon price is the endogenous dual of the emission cap; and (C) a tiered tax with Verra VM0044-compliant removal crediting (molar H/C ≤ 0.7) and allowance allocation rules. Using DOE Billion-Ton 2023 (including its USFS FIA-based forestry module) and USDA NASS data, we find that the no-policy equilibrium produces 2.30 Mt biochar/yr at a system surplus of \$708 M/yr with a net removal of 1.60 Mt CO₂e/yr. A linear carbon tax leaves this optimum unchanged (even when facilities may relocate), and net emission caps never bind, because the system is demand-constrained and intrinsically net-negative. Removal credits, by contrast, activate the \$250/t bulk-agricultural demand segment at credit prices as low as \$25/tCO₂e and mobilize essentially all near-term biomass at \$200/tCO₂e, raising surplus to \$3.36 B/yr. The endogenous marginal abatement cost traced by the cap-and-trade sweep is $\approx$ \$351–391/tCO₂e over the first binding intervals, rising to $\approx$ \$599/t, above EU ETS allowance prices and the voluntary removal-credit band; the cap’s vertex dual reproduces the marginal abatement cost to numerical tolerance, and the VM0044 eligibility gate measurably reshapes pyrolysis temperature selection. Quality-differentiated demand, which restricts the CDR premium segment to 500°C biochar, makes high-temperature pyrolysis profitable even without policy. These results quantify how carbon policy design, not merely its stringency, determines the scale, technology, and spatial structure of biochar CDR systems.

Keywords biochar supply chain; carbon dioxide removal; market clearing; inherent values; endogenous carbon price; baseline-and-credit; cap-and-trade; tiered carbon tax; VM0044; mixed-integer programming

Highlights

- The endogenous marginal abatement cost (\$351–599/tCO₂e) lies above EU ETS and voluntary removal-credit prices: gross-emission pricing is an expensive lever for a net-negative supply chain.
- Removal credits activate the bulk biochar market at \$25/tCO₂e and mobilize 99.9% of near-term Wisconsin biomass at \$200/tCO₂e.
- Net emission caps never bind for a net-negative system; negative emissions must be valued explicitly.
- The VM0044 H/C gate (0.7) excludes all torrefaction-like 300°C biochar from crediting, converting the system to 500°C pyrolysis as credit prices rise.
- Quality-differentiated demand alone makes 500°C pyrolysis (0.8 Mt/yr) profitable in the no-policy baseline.

1 Introduction

Biochar, the carbon-rich solid produced by pyrolyzing biomass under oxygen-limited conditions, is one of the most scalable engineered carbon dioxide removal (CDR) pathways available today [28, 12]. Applying biochar to soils simultaneously sequesters photosynthetically fixed carbon on centennial timescales and delivers agronomic co-benefits, including reduced nitrous oxide emissions and improved water retention [4, 21]. Recent assessments estimate that biochar could remove 0.3–2 Gt CO₂ per year globally by mid-century [29], and the 2023 Billion-Ton Report identifies agricultural residues and forestry wastes in the United States, more than a billion dry tonnes per year under mature-market scenarios, as an abundant feedstock base [24]. In parallel, voluntary carbon markets have matured to the point where biochar-based removal credits trade at \$100–250 per tonne CO₂e under methodologies such as Verra VM0044 and Puro.earth CORC, with corporate buyers purchasing hundreds of thousands of tonnes annually [5, 26].

Realizing this potential requires coordinating biomass collection, drying, pyrolysis, biochar distribution, and soil application across hundreds of facilities and counties, a classical supply chain network design problem with an unusually strong coupling between economics and carbon accounting. Unlike conventional biomass-to-energy systems, a biochar system’s dominant value stream under carbon policy is the removal credit itself, which accrues per tonne of stable carbon rather than per unit of energy sold. This coupling raises questions that existing models do not answer: “How should carbon policies be formulated inside a supply chain optimization model so that they genuinely change facility locations, technology choices, and production volumes?” and “What is the endogenous price of carbon that emerges from such a system, and how does it compare with observed market prices?”

Three strands of literature frame these questions. First, biochar techno-economic and life-cycle assessments have established the unit economics of slow pyrolysis and the permanence accounting of biochar carbon [17, 28, 15]; these studies, however, treat a single plant or a stylized regional

average and do not optimize network structure. Second, biomass and negative-emission supply chain optimization has developed spatially explicit mixed-integer models for biochar source–sink networks [22, 2], BECCS chains [14], and multi-product waste networks [18]. The market-clearing framework of Sampat et al. [18] and the spatio-temporal economic analysis of Tominac et al. [23] are particularly relevant: they show that the dual variables of mass-balance constraints in an optimization model are the “inherent values” of products at each location, i.e., equilibrium prices under perfect coordination. Third, the green supply chain literature has compared carbon taxes, cap-and-trade, and carbon credit policies in generic production and inventory networks [3, 31, 6, 7], demonstrating that a linear tax shifts the optimal solution only when emissions intensity varies endogenously across decisions; it is otherwise a constant shift of the objective.

This paper addresses the gap between these strands: no existing study (i) embeds carbon policy inside a multi-product, price-responsive supply chain network for biochar in the market-clearing/dual framework, (ii) derives the endogenous carbon price of the system as the dual of an emission constraint and benchmarks it against real market prices, and (iii) compares three canonical policy architectures (baseline-and-credit, cap-and-trade, and a tiered tax with removal crediting and allowance allocation) on identical network, data, and demand structures. We fill this gap with a case study of Wisconsin, a state whose 14 million acres of farmland, 46% forest cover, and dairy-dominated agriculture make it representative of the Midwestern biomass landscape.

The contributions of this paper are threefold. First, we formulate a multi-product biochar supply chain network design model, with 72 county nodes, 13 feedstocks, drying plus slow pyrolysis at two temperatures (300 and 500 °C), and a three-segment price-responsive biochar demand curve, in which facility locations are integer decisions and market clearing yields spatially resolved inherent values from LP duals. Second, we integrate three carbon policy architectures into this model: (A) a baseline-and-credit mechanism whose credits scale with throughput; (B) a cap-and-trade scheme in which the carbon price is the endogenous dual of the emission cap, tracing the system’s marginal abatement cost curve; and (C) a tiered carbon tax combined with Verra VM0044-compliant removal crediting (molar H/C ≤ 0.7) and grandfathering versus benchmarking allowance allocation. Third, using DOE Billion-Ton 2023 (including its USFS FIA-based forestry module) and USDA NASS 2022 Census data, we quantify how each policy reshapes facility location, technology selection, biochar production (2.25–2.85 Mt/yr across the policies studied in the near-term scenario), carbon removal (1.60–1.96 Mt CO₂e/yr), and system surplus (\$0.7–3.4 billion/yr), and we show that net emission caps alone are blind to negative emissions while removal crediting activates the low-price agricultural segment of demand at credit prices as low as \$25/tCO₂e.

The paper is structured as follows. In Section 2 we present the mathematical formulation, including the base network model and the three policy modules, together with three propositions that are proved in the Supplementary Information: the dual-to-MAC identity of the cap (Prop. 1), the demand-bound/supply-bound crossover prices that decide whether crediting upgrades the pyrolysis temperature (Prop. 2), and the dispatch-neutrality threshold of a linear emission tax in the demand-constrained regime (Prop. 3). Section 3 reports the results: baseline facility locations and economics, environmental performance, the market-clearing inherent values, the policy responses, the quality-differentiated extension, and one-at-a-time sensitivities. Section 4 discusses policy implications

against real carbon market data, and Section 5 concludes; the Wisconsin case-study data and scenario design are documented in the Supplementary Information.

2 Model Formulation

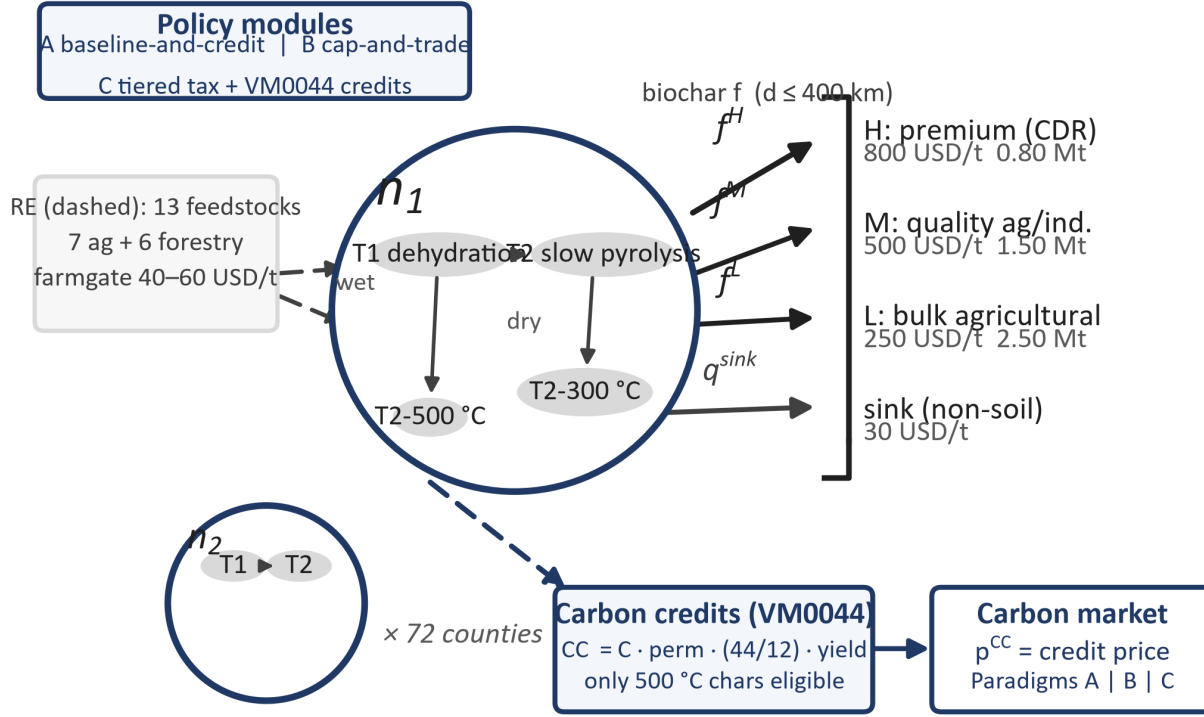


Figure 1: Supply chain superstructure: 13 feedstocks gathered at 72 county nodes, T1 dehydration, T2 slow pyrolysis at 300/500°C, biochar, carbon credits, and the price-responsive step demand. Deep-blue elements mark the carbon-accounting layer: 500°C chars (H/C 0.40–0.52) carry VM0044 credits while torrefaction-like 300°C chars (H/C 0.9–1.4) do not, and the three policy modules (A baseline-and-credit, B cap-and-trade, C tiered tax + VM0044) act on the credit and emission flows.

2.1 Problem statement

We consider a regional biochar supply chain operator (a coordinating entity, per the market-clearing paradigm of [18, 13]) that must decide where to build biomass drying and pyrolysis facilities, which feedstocks to procure from which counties, and how to distribute biochar to demand locations, in order to maximize the system-wide economic surplus (Fig. 1). The network consists of $n \in \mathcal{N}$ county nodes ($|\mathcal{N}| = 72$ for Wisconsin), $f \in \mathcal{F}$ feedstocks ($|\mathcal{F}| = 13$: corn stover, soybean straw, wheat straw, oats straw, barley straw, switchgrass, miscanthus, poplar, willow, logging residues, small-diameter trees, mill residues, and other forest waste), and a two-stage technology superstructure: T1 dehydration followed by T2 slow pyrolysis at either 300 or 500 °C. The model jointly determines (i) integer facility

locations and scale tiers, (ii) continuous material flows, and (iii) the equilibrium quantities served on a price-responsive biochar demand curve. Under carbon policy, the operator additionally earns removal credits or faces emission charges, which enter the objective as described in Section 2.5.

2.2 Base network model

The base model is a multi-product supply chain network design formulation; the complete statement (with sets, parameters, and the full constraint list) is given in the SI (Table S5). Throughout, sets are calligraphic, decision variables are lowercase, parameters are Greek or barred, and δ_{ij} denotes the great-circle distance between nodes i and j . The operator maximizes the coordinated-system surplus

$$\begin{aligned} \underset{z,x,f,\text{dem},\text{sup}}{\text{maximize}} \quad \Pi = & \underbrace{\sum_{d \in \mathcal{D}} \lambda_d^{\text{dem}} \text{dem}_d}_{\text{revenue}} - \underbrace{\sum_{s \in \mathcal{S}} \lambda_s^{\text{sup}} \text{sup}_s}_{\text{farmgate}} - \underbrace{\sum_{n,t} \omega_t (-x_{n,\text{ref}(t),t})}_{\text{OPEX}} \\ & - \underbrace{\sum_{(i,j,p) \in \mathcal{A}} (\alpha_p^{\text{var}} \delta_{ij} + \alpha_p^{\text{fix}}) f_{ijp}}_{\text{transport}} - \underbrace{\text{AF} \sum_{n,t,k} \kappa_{t,k} z_{ntk}}_{\text{annualized CAPEX}} \end{aligned} \quad (2.1a)$$

$$\text{s.t.} \quad s_{np} + p_{np} + \sum_{j:(j,n,p) \in \mathcal{A}} f_{jnp} = d_{np} + \sum_{j:(n,j,p) \in \mathcal{A}} f_{njp}, \quad (\pi_{np}) \quad n \in \mathcal{N}, p \in \mathcal{P} \quad (2.1b)$$

$$x_{n,p,t} = \gamma_{t,p} x_{n,\text{ref}(t),t}, \quad n \in \mathcal{N}, t \in \mathcal{T}, p \in \mathcal{P} \quad (2.1c)$$

$$-x_{n,\text{ref}(t),t} \leq \sum_k \bar{q}_{t,k} z_{ntk}, \quad n \in \mathcal{N}, t \in \mathcal{T} \quad (2.1d)$$

$$z_{ntk} \in \{0, \dots, 5\}, \quad 0 \leq f_{ijp} \leq \bar{f}_{ijp}, \quad \text{dem}_d \leq \bar{d}_d, \quad \text{sup}_s \leq \bar{s}_s, \quad (2.1e)$$

where $\mathcal{P}$ collects all products (wet biomass, dry biomass, biochar, carbon credits), s_{np}/d_{np} aggregate the supply and demand bids at node n , and $\gamma_{t,p}$ is the fixed-yield technology transformation. Constraint (2.1a) captures demand-side surplus from the bid stack minus farmgate, operating, transport, and annualized capital costs. Constraint (2.1b) is the node mass balance for product p ; its dual variable π_{np} is the spatial inherent value of product p at node n , the marginal cost of delivering one more unit of p at n , which the market-clearing paradigm interprets as the equilibrium price [18, 13]. Constraint (2.1c) enforces the fixed-yield technology transformations (dehydration water removal, pyrolysis mass yields, credit co-production), (2.1d) links throughput to the integer number of facilities installed at each scale tier k , and (2.1e) states integrality and the bid capacities. Transport radius limits (wet ≤ 150 km, dry ≤ 250 km, biochar ≤ 400 km) are imposed through $\bar{f}_{ijp}$, consistent with biomass-logistics cost estimates under which hauling low-density feedstocks becomes prohibitive beyond roughly 80–160 km [19]. The annualization factor is $\text{AF} = 0.1175$ ($i=10\%$, $n=20$ yr); all parameter values and data sources are tabulated in the SI.

2.3 Greenhouse gas accounting

Three GHG components enter the policy modules, all in kg CO₂e/yr. Gross emissions $E^+(x)$ comprise process emissions from collection, drying, and pyrolysis plus transport:

$$E^+(x) = \sum_{n,t} e_t^{\text{proc}} (-x_{n,\text{ref}(t),t}) + e^{\text{tr}} \sum_{(i,j,p) \in \mathcal{A}} \delta_{ij} f_{ijp}, \quad (2.2)$$

with e_t^{proc} the technology-specific process intensity per tonne of the technology's reference product (31–35 kg/t wet, i.e., 35–51 kg/t dry, for T1, 113–122 kg/t dry for T2 at 500/300°C after the syngas electricity credit) and $e^{\text{tr}} = 0.0902$ kg/t-km. Carbon removal $S(x)$ is defined as a positive physical quantity: sequestration and N_2O suppression per tonne of biochar applied:

$$S(x) = \sum_{n, t \in \mathcal{T}^{\text{pyr}}} (r_t^{\text{seq}} + r_t^{\text{N}_2\text{O}}) x_{n, \text{BC}, t}, \quad r_t^{\text{seq}} = \rho_t C_t \frac{44}{12} \times 10^3 \text{ kg/t}, \quad (2.3)$$

where C_t is the biochar carbon content, ρ_t a permanence factor ($\rho_{300} = 0.65$, $\rho_{500} = 0.80$; VM0044 v1.2 Table 3, drawn from IPCC 2019 Table 4AP.2; the 300°C value uses the lowest tabulated band, 350–450°C, and the permanence of sub-350°C chars may be lower), and $r_t^{\text{N}_2\text{O}} = 52$ kg/t from the meta-analysis of Cayuela et al. [4]. The direct atmospheric flux of the system is therefore

$$N(x) = E^+(x) - S(x), \quad (2.4)$$

which is negative (net removal) at the baseline equilibrium. This flux excludes the decomposition baseline, the emissions avoided by diverting biomass from natural decay,

$$B(x) = \sum_{f \in \mathcal{F}} b_f \sum_n (-x_{n, f, \text{T1}_f}), \quad b_f = 0.9 C_f^{\text{dry}} \frac{44}{12} \times 10^3 \text{ kg/t dry}, \quad (2.5)$$

with $b_f \approx 1350$ – 1650 kg/t dry. Throughout the paper, reported “net GHG” figures are the direct flux $N(x)$, not abatement relative to the decay counterfactual; $B(x)$ enters only the credit calculation of Paradigm A below.

2.4 Price-responsive demand and market clearing

Biochar demand follows a quality-tiered step function (bid stack) at each county, allocated in proportion to farmland area:

$$\lambda^{\text{dem}}(Q) = \begin{cases} \$800/\text{t} & Q \leq 0.8 \text{ Mt (CDR premium: BC500 only in v2b)} \\ \$500/\text{t} & 0.8 < Q \leq 2.3 \text{ Mt (quality ag / industrial media)} \\ \$250/\text{t} & 2.3 < Q \leq 4.8 \text{ Mt (bulk agricultural)} \\ \$30/\text{t} & \text{sink (construction, filtration)} \end{cases} \quad (2.6)$$

The step prices derive from the U.S. wholesale band of \$600–2,778/t and production costs of \$150–400/t (SI, demand data pack). The three soil segments sum to a finite aggregate capacity of $0.8 + 1.5 + 2.5 = 4.8$ Mt/yr, which is the ceiling on price-responsive biochar sales; the sink is unbounded (open capacity) and absorbs any residual at the low \$30/t bid. All three capacities are allocated across counties in proportion to farmland area. Because demand is price-responsive, the equilibrium quantity is endogenous; in the base model (v2) biochar is pooled, while the quality-differentiated extension (v2b, Section 2.6) splits biochar into BC300 and BC500 products with the CDR premium segment restricted to BC500.

2.5 Carbon policy modules

2.5.1 Paradigm A: baseline-and-credit

Credits accrue for the net emission reduction relative to the decomposition baseline, and because the baseline scales with throughput, the credit mechanism rewards capacity expansion, the property a linear tax lacks:

$$R(x) = B(x) - E^+(x) + S(x) \geq 0, \quad \max \Pi + \frac{p_c}{1000} R(x), \quad p_c \in \{25, 50, 100, 150, 200\} \text{ \$/tCO}_2\text{e.} \quad (2.7)$$

Implementation: carbon credit (CC) is a co-produced product with per-tonne-dry rates $\gamma_{t,28} = (b_f - e_t^{\text{proc}} + (r_t^{\text{seq}} + r_t^{\text{N}_2\text{O}})\gamma_{t,\text{BC}})/1000$, sold at the policy price p_c . Credits are computed in LCFS style for every feedstock–temperature pair, without an eligibility gate; the VM0044 H/C gate is applied only in Paradigm C2, so the two paradigms differ in both credit basis (avoided-baseline versus sequestered-carbon) and eligibility (Fig. 10).

2.5.2 Paradigm B: cap-and-trade with endogenous carbon price

B1 imposes a gross-emission cap and sweeps it to trace the system’s marginal abatement cost (MAC) curve $\Pi^*(\text{CAP})$:

$$\max \Pi \quad \text{s.t.} \quad E^+(x) \leq \text{CAP}, \quad \text{MAC} = \frac{\partial \Pi^*}{\partial \text{CAP}} \quad (2.8)$$

The endogenous-price interpretation is formalized as follows.

Proposition 1. *For the fixed-layout LP (2.1) with an added gross-emission cap $E^+(x) \leq \text{CAP}$, let μ_{CAP} be the optimal dual variable of the cap constraint under Gurobi’s sign convention for a $\leq$ constraint in a maximization. Then $\mu_{\text{CAP}} = -\partial \Pi^* / \partial \text{CAP}$: the reported dual is the negative of the marginal abatement cost, so the endogenous carbon price is $-\mu_{\text{CAP}}$.*

Proof. The identity is the standard parametric-LP sensitivity theorem applied to the cap’s right-hand side; the full argument is in the SI (Section A.9). $\square$

The proposition is verified numerically: the reported dual matches the finite-difference MAC to numerical tolerance (-374 vs. $\$351\text{--}391/\text{t}$ over the first binding intervals in the near-term scenario). We benchmark the resulting MAC against RGGI ($\$17\text{--}25/\text{t}$) and EU ETS ($\text{€}50\text{--}90/\text{t}$) allowance prices and the voluntary removal-credit band ($\$100\text{--}250/\text{t}$, VM0044/Puro). B2 replaces the constraint by a net-flux cap $E^+(x) - S(x) \leq \text{CAP}^{\text{net}}$; since the system is net-negative, the constraint never binds, demonstrating that net caps are blind to negative emissions.

2.5.3 Which resource binds: demand-bound versus supply-bound regimes

Whether a policy instrument changes the technology mix is decided by which resource binds. Define, for each feedstock f , the dry-biomass mass yields of the 300°C and 500°C pyrolysis routes γ_f^3 and γ_f^5 ($\gamma_f^3 > \gamma_f^5$), the credit rates ρ_f^3, ρ_f^5 (tCC/t dry, zero for ineligible 300°C chars under VM0044), the per-tonne dry cost ω of the full chain (farmgate, OPEX, transport, annualized CAPEX), and a biochar

market price P . The two candidate “regimes” correspond to which constraint is active in the fixed-facility LP:

- **Demand-bound** (feedstock abundant, e.g. mature-market): the supply of biochar is limited by demand, so the relevant objective margin is per tonne of biochar served. The 500°C margin premium per served tonne is

$$\Delta_f^{\text{dem}}(p_c) = \left(\frac{\rho_f^5}{\gamma_f^5} - \frac{\rho_f^3}{\gamma_f^3} \right) p_c - \omega \left(\frac{1}{\gamma_f^5} - \frac{1}{\gamma_f^3} \right), \quad (2.9)$$

independent of the market price P : a lower yield multiplies credit revenue per tonne served.

- **Supply-bound** (feedstock scarce, e.g. near-term): the supply of biochar is limited by feedstock, so the relevant margin is per tonne of dry biomass processed. The 500°C premium per dry tonne is

$$\Delta_f^{\text{sup}}(p_c) = (\gamma_f^3 - \gamma_f^5) P - (\rho_f^5 - \rho_f^3) p_c, \quad (2.10)$$

independent of the chain cost ω : the yield loss is now a revenue loss (P scales it) and the credit-rate gain is compared per dry tonne.

Proposition 2. *For a fixed feedstock f and biochar market price P : (i) in the demand-bound regime, the 500°C route becomes optimal at the unique crossing price*

$$\bar{p}_f^{\text{dem}} = \frac{\omega (1/\gamma_f^5 - 1/\gamma_f^3)}{\rho_f^5/\gamma_f^5 - \rho_f^3/\gamma_f^3}, \quad (2.11)$$

which does not depend on P and is strictly positive whenever $\rho_f^5/\gamma_f^5 > \rho_f^3/\gamma_f^3$; and (ii) in the supply-bound regime, the 500°C route becomes optimal at

$$\bar{p}_f^{\text{sup}} = \frac{(\gamma_f^3 - \gamma_f^5) P}{\rho_f^5 - \rho_f^3}, \quad (2.12)$$

which is increasing in P and bounded by the demand-segment price. Moreover, for the LCFS-style baseline-and-credit rates of Paradigm A (Table S2: feedstock averages $\bar{\gamma}^3 = 0.352$, $\bar{\gamma}^5 = 0.278$ t BC/t dry, $\bar{\rho}^3 = 1.771$, $\bar{\rho}^5 = 1.849$ tCC/t dry (the per-dry-tonne rates, whose ≈ 0.077 tCC/t dry premium is the credit-rate advantage of 500°C)) and the pooled chain cost $\omega \approx \$130/\text{t dry}$, $\bar{p}^{\text{dem}} \approx \$61/\text{tCO}_2\text{e}$ for every demand segment and $\bar{p}^{\text{sup}} \approx \$239/\text{tCO}_2\text{e}$ at $P = \$250/\text{t}$.

Proof. Equating the per-tonne margins of the two routes in each regime and solving for p_c yields the stated crossing prices; the full argument is in the SI (Section A.10). $\square$

the policy–resource interaction observed in Section 3: baseline-and-credit never upgrades the technology mix in the near-term (feedstock-scarce) scenario up to $p_c \approx \$239/\text{t}$, whereas the mature-market (demand-bound) scenario switches at $p_c \approx \$61/\text{t}$ for every demand segment; the quality tier of the premium segment ($\$800 \text{ t}^{-1}$, BC500-only) is unaffected, since it is already served by 500°C in both regimes. The proposition also explains why Paradigm C2’s VM0044 gate is so much more powerful than a flat credit price: the gate sets $\rho_f^3 = 0$, maximizing the numerator in $\bar{p}_f^{\text{sup}}$, i.e. it prices durability rather than tonnes.

2.5.4 Paradigm C: tiered tax + VM0044 eligibility + allowance allocation

C1 taxes gross emissions on a convex piecewise-linear schedule with tiers calibrated inside the feasible emission range ($T_1 = 0.4E_0^+$, $T_2 = 0.7E_0^+$; rates \$25/50/100 and \$50/100/200 per tCO₂e); the flat rate set (25/25/25) isolates the linear-tax case. To rule out the possibility that the fixed-layout approximation drives the tax results, C1 is also re-solved as a free-facility MIP (C1-free). C2 adds Verra VM0044-compliant removal crediting, operationalized through the molar H/C ≤ 0.7 eligibility gate as a proxy: only biochar with molar H/C ≤ 0.7 generates credits. Measured 300°C chars are torrefaction-like with H/C ≈ 0.9 –1.4 [16, 11], so no 300°C biochar qualifies and its credit rate is zero for every feedstock; 500°C chars have H/C ≈ 0.40 –0.52 [16, 8, 11] and qualify for all feedstocks, turning carbon crediting into a technology-selection instrument. Other VM0044 requirements (additionality demonstration, leakage accounting, monitoring and verification) are outside the model scope and their transaction costs are not modeled. C3 compares allowance allocation rules: grandfathering ($A = E_0^+$, fixed) versus benchmarking ($A = \beta q$ with $\beta = E_0^+/q_0$), with allowance trades priced at $p_{\text{allow}} \in \{30, 80\}$ \$/tCO₂e.

2.5.5 Why a linear emission tax does not change a demand-constrained system

The observed null response of the tiered tax on dispatch (Section 3.6) is not a modeling artifact but a structural property of the demand-constrained regime. Formally:

Proposition 3. *Let x^* be an optimal solution of the fixed-layout LP (2.1) and suppose the biochar mass-balance duals (inherent values) satisfy $\pi_n^{\text{BC}} \leq \lambda_{d(n)}^{\text{dem}}$ for every served demand bid $d(n)$ and $\pi_n^{\text{BC}} \geq \lambda_d^{\text{dem}}$ for every unserved bid, both with strict inequalities (the marginal served bid earns a strictly positive margin and the next, unserved, bid is strictly unprofitable; this is the demand-constrained regime with a strict gap at the margin). Then for any tax rate $\tau \in [0, \tau_{\text{max}})$ with*

$$\tau_{\text{max}} = \min_{q \in \mathcal{Q}(x^*)} \frac{\lambda_{d(q)}^{\text{dem}} - \pi_{n(q)}^{\text{BC}}}{e_q^+}, \quad (2.13)$$

where $\mathcal{Q}(x^)$ is the set of biochar supply paths that are marginally profitable at x^* ($\lambda_{d(q)}^{\text{dem}} > \pi_{n(q)}^{\text{BC}}$) and e_q^+ is the gross-emission rate of path q (kg CO₂e per tonne of biochar), the same x^* remains optimal in the taxed problem $\max \Pi(x) - \tau E^+(x)$; the tax reduces only surplus, $\Pi_\tau(x^*) = \Pi(x^*) - \tau E^+(x^*)$, and leaves the material flows, technology mix, and equilibrium quantities unchanged.*

Proof. The tax perturbs only the objective, so the optimal basis is unchanged while the reduced-cost signs of the active margin paths persist; the full argument is in the SI (Section A.11). $\square$

Intuitively, the tax rescales the objective by a constant per-unit-of-activity penalty that is uniform across the supply paths whose activity is fixed by the marginal demand bid; as long as the (gross) emission intensity of those paths is bounded relative to their surplus, the pre-tax ordering of the bid stack is preserved. In our data, the marginal price gap is $\lambda^{\text{dem}} - \pi^{\text{BC}} \approx \$155/\text{t biochar}$ (the \$500/t M bid minus the largest served-node inherent value, \$344.85/t) and the gross-emission intensity of the marginal supply path is ≈ 0.44 t CO₂e/t biochar (1,003 kt gross over 2.30 Mt served; T1 31–35 kg/t

wet, $T_2 \approx 121$ kg/t dry, plus transport), so $\tau_{\max} \approx \$350/\text{t CO}_2\text{e}$, above the top tested tier of $\$200/\text{t}$; the tax's effect within the tested range is confined to surplus reallocation.

2.6 Quality-differentiated extension (v2b)

The base model pools biochar into one product. The extension splits it into BC300 (product 27, from 300°C pyrolysis, torrefaction-like $\text{H/C} \approx 0.9\text{--}1.4$, credit-ineligible) and BC500 (product 28, from 500°C , $\text{H/C} \approx 0.40\text{--}0.52$, credit-eligible), with the CDR premium demand segment ($\$800/\text{t}$) restricted to BC500. Yields follow the V9 technology table, cross-checked against [17] and [16]: $\gamma^{\text{BC300}} = 0.34\text{--}0.38$ and $\gamma^{\text{BC500}} = 0.29\text{--}0.32$, with carbon contents 41–50% and 50–61%, respectively (the upper 500°C carbon contents are conservative relative to the 62–65% published range, Table S2). This makes the pyrolysis temperature a genuine economic decision: 500°C sacrifices yield to access the premium market and higher removal credits.

2.7 Solution strategy

Facility-location decisions are integer, so Steps 1 (MIP) and the policy sweeps use Gurobi 13 with a 1–3% optimality gap, 600–1,800 s time limits, and feasibility-focused branching (MIPFocus=1) with 60 s of zero-objective heuristics; successive policy prices are warm-started from the previous facility layout. All dual-based analyses (inherent values, GHG shadow prices, endogenous carbon prices) are performed on LPs with the facility layout fixed and solved with dual simplex, so that reported duals are exact vertex shadow prices; interior-point (barrier) duals of the cap constraint are not vertex shadow prices on this LP and are not used. The GHG LP (Step 3) pins demand quantities to the Step-2 equilibrium, so its duals are marginal GHG values at the market equilibrium rather than at the GHG-minimal dispatch. The near-term scenario (6.97 Mt dry biomass) serves as the baseline and the mature-market medium scenario (13.53 Mt dry) as sensitivity. Full reproducibility scripts and data are provided in the SI.

Assumption 1. *Throughout our study we assume that strong duality holds for the fixed-layout LPs, so that all reported dual variables are unique vertex shadow prices of the corresponding constraints.*

3 Results

3.1 Baseline facility locations and economics

Without carbon policy, the optimal network serves exactly the two highest-value demand segments and nothing else (Table 1; see also Fig. 3 for the market-clearing geometry). The equilibrium quantity is 2.30 Mt of biochar per year: the H segment ($\$800/\text{t}$, 0.80 Mt) and M segment ($\$500/\text{t}$, 1.50 Mt) are fully served, while the L segment ($\$250/\text{t}$, 2.5 Mt) remains unserved because the marginal cost of producing and delivering an additional tonne of biochar exceeds $\$250/\text{t}$. The system surplus is $\$708$ M/yr in the near-term scenario and $\$736$ M/yr under mature-market supply; wet biomass consumption reaches 6.9 Mt/yr (77% of the near-term potential). Corn stover is the dominant feedstock (5.9 Mt

wet, fully utilized), supplemented by soybean straw and logging residues. The optimal facility layout concentrates plants in the central and southern corn belt (Fig. 2); the cost structure is shown in Fig. 3.

Table 1: Baseline (S0) system summary, pooled-biochar model (v2); the quality-differentiated model (v2b) is reported separately in Table 3.

	near-term	mature-market medium
System surplus (M\$/yr)	708.3	735.7
Biochar (Mt/yr)	2.30	2.30
H / M / L served (Mt)	0.80 / 1.50 / 0	0.80 / 1.50 / 0
Wet biomass (Mt/yr)	6.91	6.60
Net GHG (Mt CO ₂ e/yr)	−1.60	−1.65
Facilities (T1 / T2-300 / T2-500)	78 / 38 / 0	29 / 18 / 0

Optimal facility locations (S0 baseline, near-term)

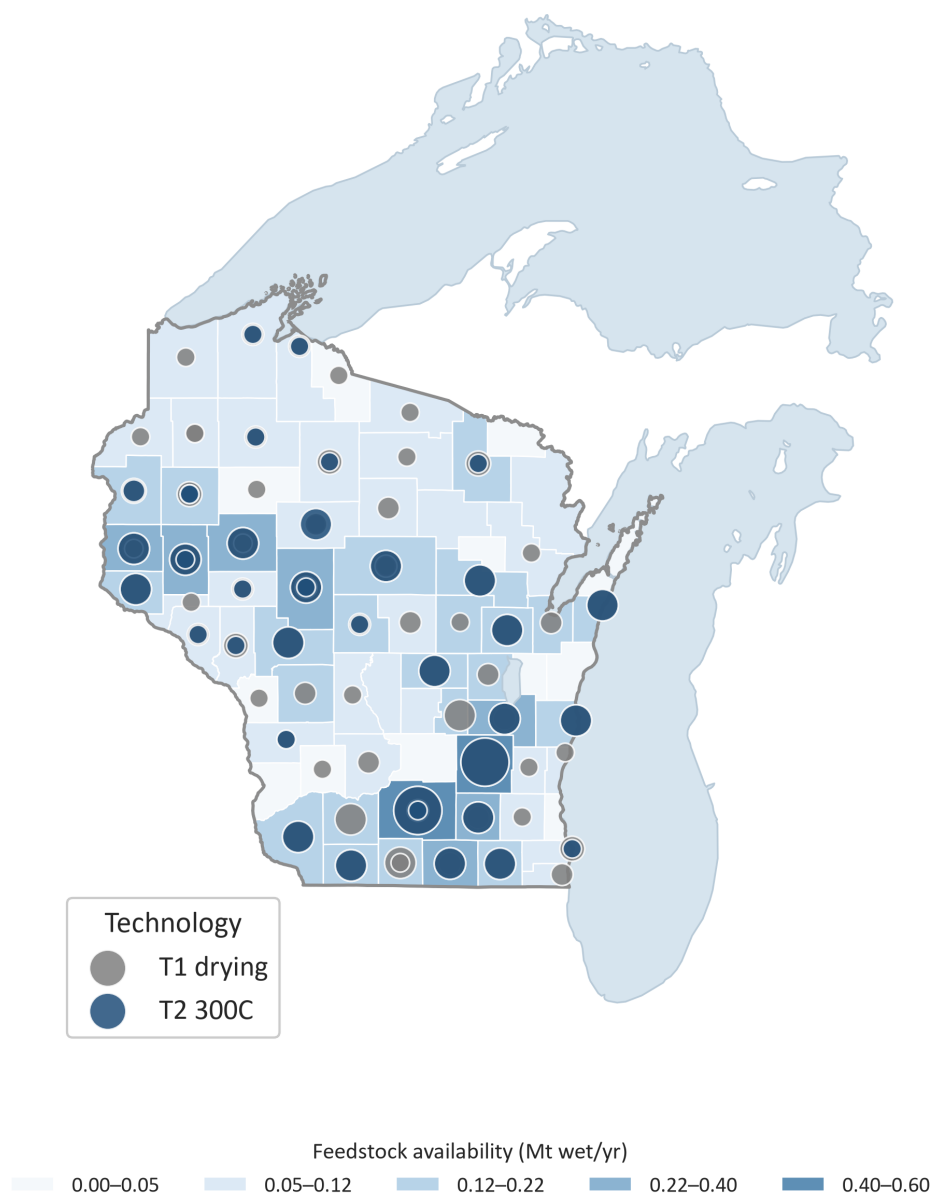


Figure 2: Optimal facility locations and scale tiers (near-term, no policy). Gray markers: T1 dehydration; deep-blue markers: T2 pyrolysis (300°C); marker size proportional to installed scale tier. Facilities concentrate in the central and southern corn belt, co-locating drying and pyrolysis at feedstock-rich counties so that wet-biomass haulage stays short, the spatial signature of a supply-tight, feedstock-driven network.

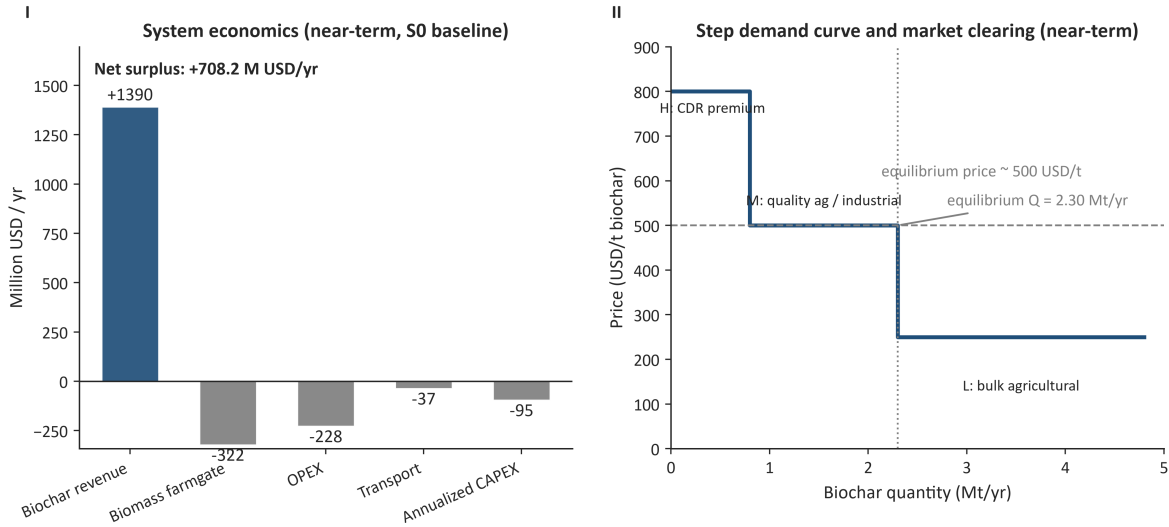


Figure 3: System economics and market clearing (near-term). (I) Revenue and cost decomposition: biochar revenue (deep blue) against farmgate, OPEX, transport, and annualized capital costs (gray), leaving a net surplus of \$708 M/yr. (II) Step demand curve and equilibrium: the marginal supply curve crosses demand inside the \$500/t M segment, so H and M are fully served (0.80 and 1.50 Mt) while the \$250/t L segment and the \$30/t sink stay unserved; this is the demand-constrained regime that policy instruments must overcome.

3.2 Environmental performance

The direct atmospheric flux $N(x) = E^+(x) - S(x)$ is -1.60 Mt CO₂e/yr (process +979 kt, transport +24 kt, carbon removal 2,598 kt) in near-term and -1.65 Mt (process +967 kt, transport +16 kt, removal 2,631 kt) under mature-market supply; it excludes the decomposition baseline $B(x) \approx 1.35\text{--}1.65$ t CO₂e/t dry, which enters only the Paradigm A credit calculation (Section 2.3). Process emissions dominate the gross flux because drying and pyrolysis are energy-intensive per delivered tonne, while the sequestration term scales directly with biochar output; the system is therefore net-negative in both scenarios without any policy intervention, and its environmental profile is a co-product of the economic equilibrium rather than a separate objective.

3.3 Value of biochar and biomass: market-clearing inherent values

The duals of the node mass balances in the fixed-layout LP (Fig. 4) reveal the spatial inherent values of biochar and biomass. In the near-term scenario the inherent value of biochar ranges \$305–\$345/t (mean \$324/t), i.e., the marginal value of an additional tonne of biochar available at a county equals its marginal production, delivery, and capacity-scarcity cost at the fixed baseline layout; under mature-market supply this falls to \$260–\$288/t (mean \$275/t) because abundant low-cost feedstocks (miscanthus, energy crops) shrink the scarcity rent. Wet biomass inherent values are far below farmgate prices (\$0–\$77/t, mean \$9/t in near-term, Fig. 4), confirming that the system is demand-constrained: additional biomass supply is nearly worthless without additional biochar demand. Following the market-clearing paradigm, these duals are interpreted as equilibrium prices under a coordinating entity; they are marginal values conditional on the baseline facility layout, not prices

from a decentralized Nash market (Section 4.6). This demand-constrained structure is exactly the regime in which a linear carbon tax fails to move the optimum (cf. [3]).

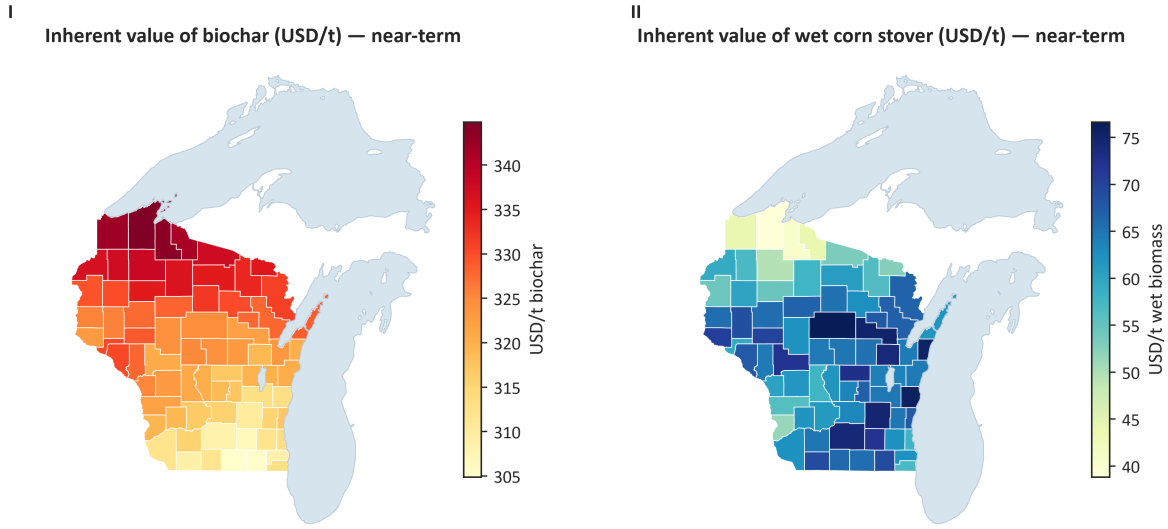


Figure 4: Inherent values at each county (duals of the node mass balances; near-term scenario). (I) Biochar inherent value \$305–345/t: the marginal value of an additional biochar tonne available at a county. (II) Wet corn stover inherent value \$39–77/t, the most valuable biomass series; other wet feedstocks carry near-zero values (mean \$9/t across all wet products), confirming the demand-constrained regime in which marginal biomass supply is nearly worthless without additional biochar demand.

3.4 Paradigm A: baseline-and-credit

Credits that scale with throughput transform the system (Fig. 5). At $p_c = \$25/\text{tCO}_2\text{e}$ the L segment activates (0.42 Mt served) and production rises to 2.72 Mt; at $p_c = \$200/\text{tCO}_2\text{e}$ the system saturates its feedstock potential (2.85 Mt, i.e., essentially 100% of near-term biomass) and surplus reaches \$3.36 B/yr. In the mature-market scenario the response is far larger: 4.79 Mt at $p_c = \$25$ and ≈ 4.80 Mt at $p_c \geq \$50$, with the L segment fully served from $p_c = \$25$ and production then pinned at the feedstock ceiling; surplus reaches \$5.87 B/yr at $p_c = \$200$. All sweep points solve at a 0.5% gap target with 600 s time limits (the $p_c = 25/50$ points re-solved with 1,800 s limits, improving the incumbents); reported gaps are 0.95–1.9% (near-term) and 0.5–3.2% (mature-market, SI Table S3), and TIME_LIMIT points are feasible incumbents rather than proven optima, so their production and surplus figures are lower bounds on the true policy response. A dedicated 1,800 s re-solve of the mature $p_c = \$100$ point at a 0.1% gap target (archived as `policy_A_recert_v2.csv`) raised its 500°C share from 1.75 to 2.79 Mt at a 1.9% gap, matching the $p_c = \$150$ –\$200 points (2.81 Mt at 1.0–1.7% gaps). Whether credits change the system’s technology mix is decided by which resource binds, not by the credit price alone (Fig. 10; Proposition 2 formalizes the crossover prices in the demand-bound and supply-bound regimes). In the feedstock-scarce near-term scenario the 300°C chain remains optimal for all biochar demand segments up to $p_c \approx \$239/\text{t}$, because its yield advantage (0.352 versus 0.278 t/t, feedstock averages) outweighs the modest credit-rate premium of 500°C biochar ($\approx 0.077 \text{ tCC}/\text{t dry}$); the binding resource is dry

feedstock, so per-dry-tonne margins rule. In the feedstock-abundant mature-market scenario the binding resource is biochar demand, and 500°C's lower yield multiplies credit revenue per tonne of biochar served: 500°C overtakes 300°C at $p_c \approx \$61/\text{t}$ for every demand segment, and the sweep routes ≈ 2.8 Mt of the 4.8 Mt through 500°C at $p_c = \$100\text{--}\200 . Baseline-and-credit therefore scales the system in both regimes, but induces a quality upgrade only where feedstock is abundant, a policy–resource interaction that a single-regime analysis cannot reveal. These credits are LCFS-style baseline-and-credit, granted to every feedstock–temperature pair without an eligibility gate; the VM0044 H/C gate is applied only in Paradigm C2 (Section 3.6), so Paradigms A and C differ in credit basis as well as eligibility (Fig. 10). The direct atmospheric flux $N(x) = E^+(x) - S(x)$ deepens from -1.60 to -1.96 Mt CO₂e/yr as credit prices rise, i.e., net removal grows with mobilization.

3.5 Paradigm B: cap-and-trade with endogenous carbon price

The gross-emission cap sweep traces the system's marginal abatement cost curve (Fig. 6). Tightening the cap from the baseline 1,003 kt to 501 kt reduces surplus from \$708 M to \$451 M/yr and biochar from 2.30 to 1.19 Mt (mature-market: 983→492 kt, \$736→\$458 M). The finite-difference marginal abatement cost is $\approx \$351\text{--}\$391/\text{t}$ CO₂e over the first binding intervals and rises to $\approx \$599/\text{t}$ as the cap tightens (mature-market $\approx \$468\text{--}\$484 \rightarrow \$603/\text{t}$), above both EU ETS (€50–90/t) and the voluntary removal-credit band (\$100–250/t). The cap's vertex dual reproduces these values exactly: with Gurobi's sign convention the reported dual is the negative of the marginal value, i.e., $-\$374/\text{t}$ at the first binding cap (near-term), inside the \$351–391/t finite-difference band of the first binding intervals, validating the endogenous-price interpretation. Gross emissions are intrinsic to drying and pyrolysis, so abating them means forgoing biochar sales, whereas carbon removal is a co-benefit of production; pricing gross emissions is therefore an expensive lever for this system. The net-flux cap experiment completes the picture: for $\text{CAP}^{\text{net}} \in [-1,000, +1,000]$ kt the constraint $E^+(x) - S(x) \leq \text{CAP}^{\text{net}}$ never binds ($\pi = 0$) and surplus stays at \$708 M/yr, while a cap of $-2,000$ kt is infeasible (the GHG-minimal dispatch already exceeds it). Because the system is intrinsically net-negative, a net cap provides zero incentive; negative emissions must be valued explicitly through credits.

3.6 Paradigm C: tiered tax, VM0044 eligibility, and allocation

The tiered tax (C1) reduces surplus (708 to 653 M\$/yr at rates 25/50/100) but does not alter the fixed-facility dispatch, consistent with the demand-constrained regime (Proposition 3 establishes that a linear tax cannot change the dispatch when demand is marginally binding). To rule out the possibility that this null response is an artifact of the fixed layout, we re-solve C1 as a free-facility MIP (C1-free): the optimizer finds a layout equivalent to the baseline within the MIP gap at every rate set tested (Table 2; 75–80 T1 and 38–40 T2 units versus 78 and 38 at baseline, with gross emissions within 1% of the baseline dispatch). The free-facility surplus (653.3 M\$/yr) matches the fixed-layout LP (653.2 M\$/yr) to within the MIP gap, and the same holds for the steeper (598.3 vs 598.0 M\$/yr) and linear (683.2 vs 683.2 M\$/yr) schedules, confirming that the tax's ineffectiveness is economic (demand, not emissions, binds) rather than a modeling artifact. Adding VM0044 removal crediting with free facilities (C2) changes technology selection directly, because the eligibility gate makes no 300°C biochar

credit-eligible: measured 300°C chars are torrefaction-like ($H/C \approx 0.9\text{--}1.4$), while 500°C chars ($H/C \approx 0.40\text{--}0.52$) qualify for every feedstock. At $p_c = \$50$ the near-term system is unchanged (2.30 Mt, all 300°C), but the mature-market system already routes 0.50 Mt to 500°C; at $p_c = \$100$ the conversion is complete in both scenarios: 2.22 Mt (near-term) and 2.71 Mt (mature-market), entirely 500°C. At $p_c = \$200$ production is 2.25 Mt (near-term, surplus \$1,141 M/yr, 3.64 Mt of removal credits) and 4.25 Mt (mature-market, surplus \$1,501 M/yr, 7.03 Mt credits), entirely 500°C (Fig. 7). Two features of this result are notable. First, the gate never subsidizes low-durability biochar: every credited tonne satisfies $H/C \leq 0.7$, and the credits purchase durability rather than tonnes. The near-term system’s 2.25 Mt falls below the 2.30 Mt H+M demand target because 500°C yields less char per dry tonne, and the mature-market 4.25 Mt remains below the 4.80 Mt achieved under ungated baseline-and-credit crediting (Section 3.4). Second, full conversion raises gross emissions (E^+ grows from 1,003 to 1,204 kt near-term and from 983 to 2,239 kt mature-market) because each delivered biochar tonne now requires more drying and pyrolysis throughput, exactly the interaction a tiered emission tax is designed to price. The allowance-allocation comparison (C3) shows no difference between grandfathering and benchmarking under fixed facilities: with the layout fixed, gross emissions are pinned by the demand-constrained dispatch, the allowance budget is not a binding decision, and the allocation rule merely relabels a fixed revenue stream. Allocation design therefore matters only when facility investment is endogenous, an avenue for the multi-period extension. Figure 8 consolidates the production, surplus, and price responses across the three paradigms.

Table 2: Tiered tax with free facilities (C1-free): the free-facility MIP finds a layout equivalent to the baseline within the MIP gap at every rate set, confirming that the null tax response is driven by the demand-constrained regime rather than the fixed-layout approximation (fixed-layout LP surplus in parentheses).

rates (r1,r2,r3)	near-term		mature-market medium	
	surplus (M\$)	E^+ (kt)	surplus (M\$)	E^+ (kt)
25/50/100	653.3 (653.2)	1,003	683.1 (681.7)	985
50/100/200	598.3 (598.0)	998	629.3 (627.6)	983
25/25/25 (linear)	683.2 (683.2)	1,002	712.9 (711.2)	983

3.7 Quality-differentiated biochar (v2b)

Splitting biochar into BC300 and BC500 with the \$800/t CDR premium segment restricted to BC500 changes the baseline itself: the optimal no-policy network builds 14 T2-500°C facilities and produces 0.80 Mt of BC500, exactly the premium segment, alongside 1.50 Mt of BC300 (Fig. 9, Table 3); the premium market alone justifies high-temperature pyrolysis. The v2b no-policy surplus (\$636 M/yr near-term, \$674 M/yr mature-market) is lower than the pooled-baseline surplus (\$708 M/yr) because serving the premium segment with BC500 incurs the higher energy cost and lower yield of 500°C pyrolysis; the two models therefore define different counterfactuals, and cross-model comparisons should focus on production and technology mix rather than surplus levels. The response to baseline-

and-credit prices follows the two-regime logic of the pooled model. In the feedstock-scarce near-term scenario, the bulk expansion is served entirely by BC300 (1.50→1.84 Mt), and BC500 stays pinned at the 0.80 Mt premium segment, because per-dry-tonne margins keep 300°C optimal up to $p_c \approx \$239/\text{t}$ (Fig. 10). In the feedstock-abundant mature-market scenario the demand-bound regime takes over: BC500 rises from 0.80 to 2.74–2.82 Mt at $p_c = \$100\text{--}\200 , roughly 60% of the 4.8 Mt total, as 500°C's lower yield multiplies credit revenue per tonne of biochar served above $p_c \approx \$61/\text{t}$. Technology selection is therefore jointly driven by the credit price and the resource regime, a policy–resource interaction that a pooled-biochar model without quality differentiation cannot fully reveal, since it also cannot see the premium-segment floor on BC500.

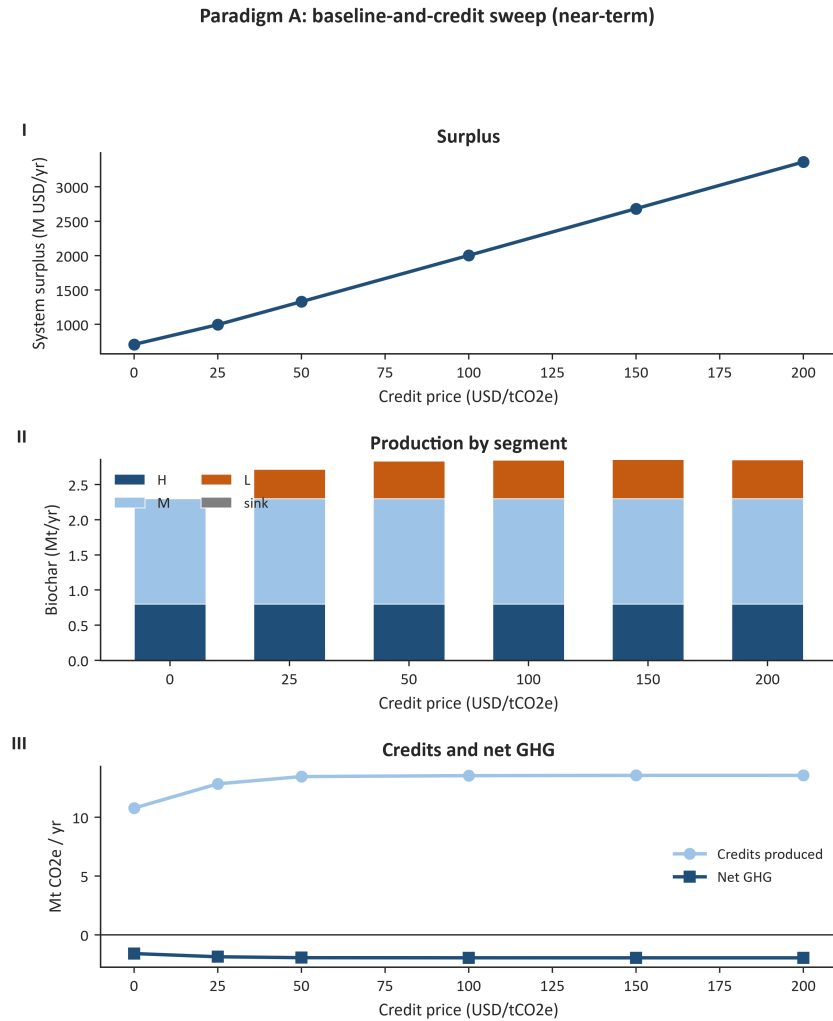


Figure 5: Paradigm A (baseline-and-credit): surplus, production by demand segment, and credits/net GHG as functions of the credit price. Throughput- scaled credits move the equilibrium: the \$250/t L segment activates at \$25/tCO₂e and essentially all near-term feedstock is mobilized at \$200/tCO₂e, tripling surplus; removal crediting, not emission pricing, is the expansion lever.

Paradigm B: cap-and-trade with endogenous price (near-term)

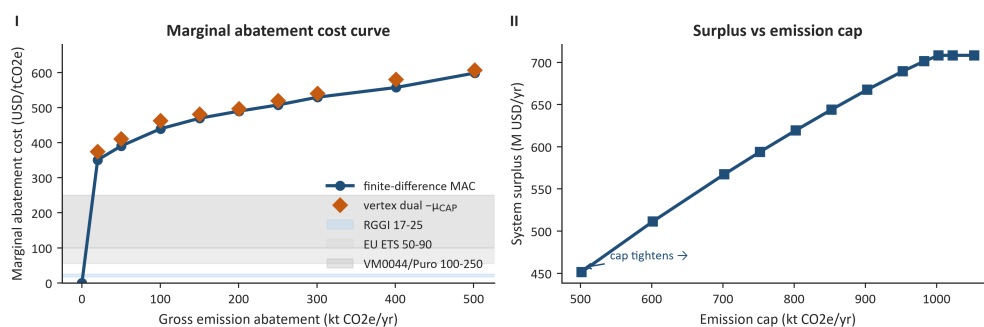


Figure 6: Paradigm B: (I) marginal abatement cost curve with benchmark market price bands (RGGI, EU ETS, VM0044/Puro); (II) surplus versus the gross emission cap. The endogenous carbon price starts at $\approx$ \$351–391/t CO₂e and rises to $\approx$ \$599/t, above every existing allowance and voluntary-credit price; gross-emission pricing is an expensive, quantity-reducing lever for this net-negative system.

Paradigm C: tiered tax (25/50/100) + VM0044 credit (near-term)

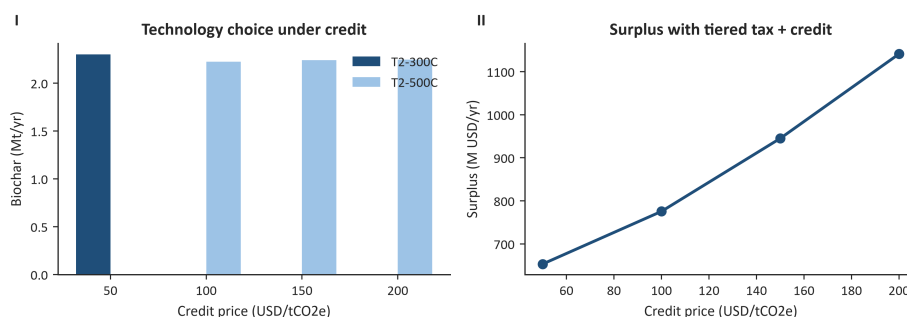


Figure 7: Paradigm C (tiered tax + VM0044 credit): technology choice 300 vs 500°C (I) and surplus (II). The tiered tax leaves the dispatch unchanged, but VM0044 crediting, whose $H/C \leq 0.7$ gate excludes every torrefaction-like 300°C char, converts the system to 500°C pyrolysis as credit prices rise: partial at \$50/t (mature-market), complete at \$100/t, and full conversion (2.25 Mt near-term, 4.25 Mt mature-market) at \$200/t.

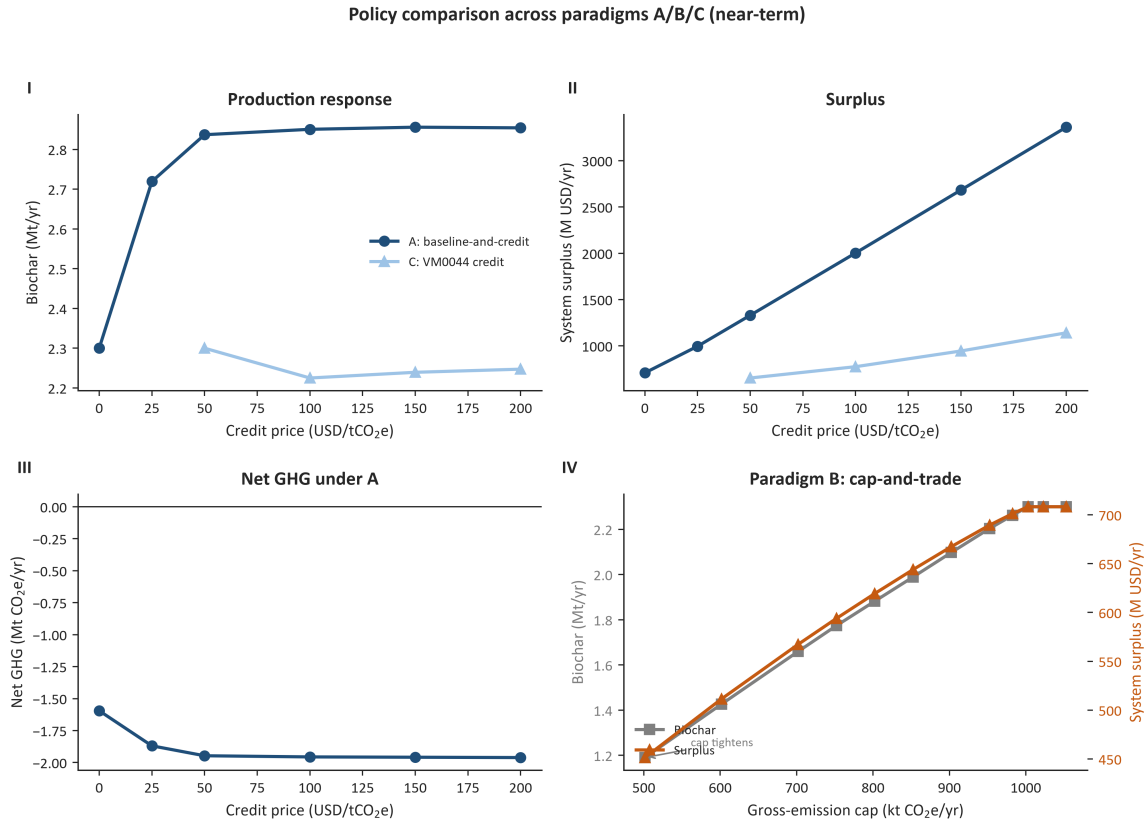


Figure 8: Policy comparison across paradigms A/B/C: (I) production, (II) surplus, (III) net GHG, (IV) endogenous versus market carbon prices. Paradigms A and C2 are free-facility investment responses while B and C1 are fixed-layout dispatch responses, and the panel axes differ by design (credit price \$/tCO₂e for A/C, cap in kt for B); the comparison is therefore of policy mechanisms, not of like-for-like stringency. Only throughput-scaled removal crediting (A) and the gated VM0044 credit (C2) expand the system, while the tiered tax (C1) is inert and the gross-emission cap (B) shrinks it at a carbon price above every real market price.

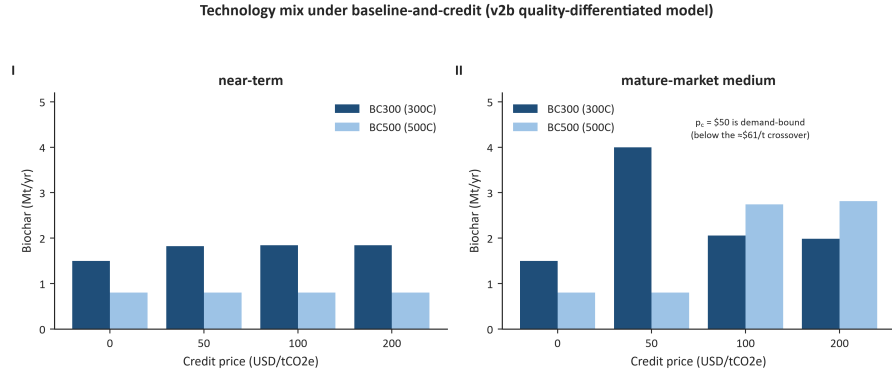


Figure 9: Quality-differentiated model: BC300/BC500 technology mix under baseline-and-credit in both supply scenarios. Near-term, the premium segment keeps BC500 at exactly 0.80 Mt while expansion runs on BC300; mature-market, 500°C's lower yield multiplies credit revenue per tonne served and BC500 captures 2.74–2.81 Mt, the demand-bound crossover at $p_c \approx \$61/t$.

Table 3: Quality-differentiated model (v2b): technology mix under baseline-and-credit. BC300/BC500 in Mt/yr.

p_c (\$/tCO ₂ e)	near-term				mature-market medium			
	BC300	BC500	total	profit	BC300	BC500	total	profit
0	1.50	0.80	2.30	636	1.50	0.80	2.30	675
50	1.82	0.80	2.62	1,283	4.00	0.80	4.80	1,729
100	1.84	0.80	2.64	1,972	2.06	2.74	4.80	3,097
200	1.84	0.80	2.64	3,350	1.99	2.81	4.80	5,861

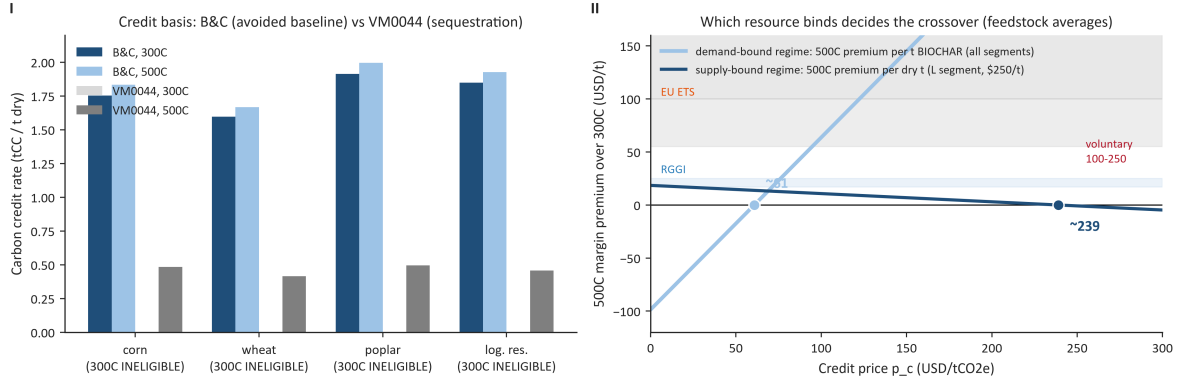


Figure 10: The credit basis and the two-regime 300/500°C technology crossover. (I) Carbon credit rates per tonne dry biomass under the two credit bases (baseline-and-credit, LCFS-style, $B - E^+ - S$, versus VM0044, sequestration-based and H/C-gated) for representative feedstocks and both pyrolysis temperatures; note the zero VM0044 rate of every torrefaction-like 300°C char. (II) The 500°C margin premium over 300°C against the credit price in the two resource regimes (feedstock averages). Demand-bound regime (feedstock abundant): the binding resource is biochar demand, and 500°C’s lower yield multiplies credits per tonne served, with crossover at $\approx \$61/\text{tCO}_2\text{e}$ for every segment. Supply-bound regime (feedstock scarce): per-dry-tonne margins rule, with crossover at $\approx \$239/\text{tCO}_2\text{e}$ for the bulk segment. Shaded bands: RGGI, EU ETS, voluntary-market credit prices.

3.8 One-at-a-time sensitivity

The baseline solution is far more sensitive to demand-side than to supply-side parameters (S-Figure S1). Demand price $\pm 20\%$ moves the system surplus by $\pm 39\%$ (from \$430 M to \$986 M/yr) while leaving the quantity at the demand-constrained equilibrium; a 20% reduction in demand capacity reduces surplus by 21% and quantity by 20%. In contrast, farmgate prices $\pm 20\%$ change surplus by only $\pm 9\%$ and transport costs $\pm 20\%$ by about $\pm 1\%$; the sink price has no effect because the sink is not activated. A 40% reduction in feedstock availability (a proxy for realistic field-collection losses) reduces production to 1.48 Mt and surplus by 29% in the near-term scenario (from 708.3 to 505.9 M\$/yr), but only by 4% in the mature-market scenario: the system is demand-constrained at the aggregate level, but its preferred feedstock mix (near-term corn stover) is supply-tight, so availability shocks matter through the composition of supply even when total demand binds. These one-at-a-time perturbations hold the baseline facility layout fixed, so they measure dispatch-level responses; layout-level responses are captured separately by the free-facility policy sweeps (Paradigms A, C1-free, and C2). The asymmetry confirms that demand-side instruments (adoption support, removal crediting) are the effective levers for expanding biochar CDR in Wisconsin, while feedstock-availability risk concentrates in the corn-stover supply chain. Two policy-relevant parameter sensitivities complete the picture. First, the decomposition-baseline assumption (0.9 decay fraction) enters only Paradigm A’s credit rates: halving it to 0.5 lowers the \$200/t surplus from \$3.36 to \$2.33 B/yr (-31%) in the near-term and from \$5.87 to \$3.81 B/yr (-35%) mature-market, while mobilized production stays near the feedstock potential; the qualitative response is robust, but the surplus level scales roughly one-for-one with the baseline

assumption. Second, the premium-segment capacity directly sets the scale of 500°C deployment: halving H to 0.4 Mt halves BC500 to 0.40 Mt (total production 1.90 Mt), and enlarging H by 50% (1.2 Mt) raises BC500 to 1.20 Mt and total production to 2.52 Mt near-term (2.70 Mt mature); the premium market, not production cost, is the binding driver of high-temperature pyrolysis.

4 Policy Discussion

4.1 Why a linear carbon tax fails and what works instead

The baseline system is demand-constrained: at the equilibrium, the binding constraints are the biochar demand segments, not biomass availability. In this regime a linear tax $\tau E^+(x)$ on gross emissions is a near-constant shift of the objective (gross emissions change little across feasible dispatches), so the primal optimum does not move, precisely the failure mode characterized by [3]. Proposition 3 states the boundary condition precisely: the tax is dispatch-neutral up to a threshold whose denominator is the marginal price gap divided by the path emission intensity, which we compute at $\approx \$350/\text{t CO}_2\text{e}$, above every tested rate ($\$25\text{--}200/\text{t}$) but close enough that substantially higher taxes could bite. The free-facility re-solve (C1-free, Section 3.6) confirms that this null response is economic rather than an artifact of the fixed layout. The claim’s boundary condition is equally important: it holds because demand binds; in a supply-constrained variant of the system (e.g., feedstock scarcity binding before demand) a tax would bite, and the statement should not be read as tax-neutrality of biochar systems in general. Our formulation shows that policy instruments which scale with throughput (baseline-and-credit) or which act as constraints (gross emission caps) do change the solution: credits activate the $\$250/\text{t}$ bulk-agricultural demand segment at $p_c = \$25/\text{tCO}_2\text{e}$ and mobilize essentially all near-term biomass at $\$200/\text{tCO}_2\text{e}$, while caps reshape production along the marginal abatement cost curve. The net-flux cap experiment completes the picture: because the system is intrinsically net-negative, CAP^{net} never binds and provides zero incentive. Policy design for negative-emission supply chains must therefore explicitly value removal, not merely penalize emissions.

4.2 Endogenous carbon prices versus real markets

The endogenous marginal abatement cost traced by the gross-emission cap sweep is $\approx \$351\text{--}391/\text{t CO}_2\text{e}$ over the first binding intervals and rises to $\approx \$599/\text{t}$ as the cap tightens; the cap’s vertex dual reproduces it exactly (the reported value is the negative of the marginal value under Gurobi’s sign convention), validating the endogenous-price interpretation on a full supply chain network. This places the Wisconsin biochar system’s gross-emission abatement above RGGI allowance prices ($\$17\text{--}25/\text{t}$), EU ETS prices ($\text{€}50\text{--}90/\text{t}$), and the voluntary removal credit band ($\$100\text{--}250/\text{tCO}_2\text{e}$ for VM0044/Puro.earth credits). Two implications follow. First, at any observed carbon price, a cap-and-trade program alone would induce little gross abatement in this system, because gross emissions are intrinsic to drying and pyrolysis and abating them means forgoing biochar sales; reaching the $\$800/\text{t}$ -market equilibrium required no policy at all, but expanding to the bulk agricultural segment requires credit prices above roughly $\$25/\text{tCO}_2\text{e}$, well within observed voluntary-market prices. Second, the

comparison of the system’s endogenous abatement cost with administered market prices shows that removal crediting, not gross-emission pricing, is the mechanism that can profitably expand biochar CDR, a quantitative argument for designing negative-emission policies around the removal, rather than around the gross emissions, of such systems.

4.3 Correcting a common policy misattribution: 45Q

We note explicitly that the U.S. 45Q tax credit, frequently cited in biochar investment documents, does not apply to biochar. 45Q credits captured CO₂ streams (point-source CCS and DAC); biochar carbon is sequestration, not capture, and falls outside the statute. Biochar removal credits are instead issued under voluntary methodologies (Verra VM0044, which requires molar H/C ≤ 0.7 and centennial durability, and Puro.earth CORC); VM0044 credits were endorsed by the ICVCM as high-integrity in 2025 [1]. The policy landscape is, however, in motion: legislation under consideration in the 119th Congress (e.g., the Wildfire Reduction and Carbon Removal Act of 2025, S. 1842) would create a federal biochar carbon-removal incentive and reform 45Q, which could materially change the economics quantified here [25]. Our Paradigm C2 operationalizes the VM0044 gate with literature-consistent H/C values: 300°C biochar is torrefaction-like (H/C ≈ 0.9 –1.4) and earns no credits for any feedstock, while all 500°C chars (H/C ≈ 0.40 –0.52) qualify. The model responds by converting the system to 500°C pyrolysis as credit prices rise: a partial switch already at \$50/t in the mature-market scenario, complete at \$100/t, and full conversion (2.25 Mt near-term, 4.25 Mt mature-market) at \$200/t; the conversion buys durability at the cost of quantity: near-term production falls slightly below the 2.30 Mt H+M demand target because 500°C yields less char per dry tonne. The gate switches technology without ever subsidizing low-durability biochar, an example of policy detail directly reshaping technology selection.

4.4 Technology selection and quality differentiation

Two mechanisms drive pyrolysis temperature choice, and which one binds is itself the finding. First, when dry feedstock is scarce (near-term), per-dry-tonne margins rule: 300°C dominates for every demand segment up to $p_c \approx \$239/\text{t}$, because its yield advantage (0.352 vs 0.278 t/t) outweighs the modest credit-rate premium of 500°C biochar. Second, when feedstock is abundant (mature-market), the binding resource is biochar demand, and 500°C’s lower yield becomes an asset: each tonne of biochar served requires more dry tonnes, each earning credits, so 500°C overtakes 300°C at $p_c \approx \$61/\text{t}$ for every segment; the model switches ≈ 2.8 Mt of production at \$100–\$200/t (the \$100/t point from a dedicated 1,800 s re-solve at a 1.9% gap, archived as `policy_A_recert_v2.csv`). Third, quality-differentiated demand (v2b) adds a structural floor: the \$800/t premium segment is BC500-only, so the no-policy baseline already builds 0.80 Mt/yr of BC500 capacity, and varying the premium capacity moves BC500 one-for-one (0.40 Mt at half capacity, 1.20 Mt at 1.5 $\times$). The market structure (whether buyers differentiate biochar by durability) therefore matters as much as the carbon price, and the resource regime decides which margin arithmetic applies.

4.5 Implications for Wisconsin and the Midwest

The analysis yields three region-specific messages; magnitudes are calibrated to Wisconsin, while the mechanisms should transfer qualitatively across the Upper Midwest (Section 4.6). (i) Demand, not feedstock, is the binding constraint: Wisconsin’s near-term biomass could support roughly double the equilibrium biochar production if the \$250/t bulk market is activated; policies that subsidize farmer adoption (or that value carbon at voluntary-market rates) expand the system more effectively than supply-side measures. (ii) Facility locations concentrate in the corn belt and shift with supply scenarios: mature-market supply lowers biochar’s spatial inherent value from \$324 to \$275/t, flattening location rents. (iii) The dairy-dominated waste streams (manure, paper-mill sludge) that characterize Wisconsin are outside this pyrolysis-focused model but form a natural co-location layer for anaerobic-digestion integrated bioeconomies, an explicit direction for the multi-pathway extension.

4.6 Limitations

Several assumptions bound the interpretation of these results. (i) *Market clearing*. The model solves a single coordinated system optimum of Sampat et al. [18]; the dual-based “equilibrium prices” are marginal values under perfect coordination, not outcomes of a decentralized Nash market, so magnitudes (especially policy responses) should be read as coordination-consistent benchmarks. (ii) *Technology space*. Only slow pyrolysis at two temperatures (300/500°C) is modeled; intermediate temperatures and competing conversion routes (gasification, hydrothermal carbonization) are excluded, so the technology-choice conclusions are discrete comparisons of two points, not a continuous design response. (iii) *Transport radii*. Hard distance limits (150/250/400 km) approximate continuous logistics costs; they follow literature cost benchmarks but truncate the feasible region. (iv) *Feedstock availability*. BT23 tonnages are treated as fully collectable; field-level collection losses and residue retention requirements are not modeled. A –40% availability sensitivity costs 29% of near-term surplus (the preferred corn-stover mix is supply-tight) but only 4% in the mature scenario, so the assumption matters mostly for near-term deployment. (v) *GHG parameters*. The decomposition fraction (0.9), the permanence factors (0.65/0.80, now aligned with VM0044 v1.2 Table 3), the literature-based char H/C values (Table S2; the eligibility outcome is inert to ± 0.1 H/C and the first 500°C flip needs +0.25, SI Section on eligibility sensitivity), and the N₂O meta-analysis mean (52 kg/t) are literature defaults; the decomposition fraction is flagged for local calibration, and a sensitivity halving it to 0.5 lowers the \$200/t Paradigm A surplus by 31–35% (Section 3.8); the qualitative response is robust, the surplus level is not. (vi) *VM0044 proxy*. Eligibility is operationalized through the molar H/C gate only; additionality, leakage, and monitoring/verification costs are outside the model scope. (vii) *Demand*. Demand segments are exogenous step functions calibrated to market evidence rather than estimated elasticities. (viii) *Static structure*. The model is single-year with no seasonal inventory; the allocation-design result (C3) and facility-path dependencies point to a multi-period extension. (ix) *Solver gaps*. Policy sweeps are reported at 0.5–4.7% MIP optimality gaps (Table S3); the points with the largest gaps are re-solved with 1,800 s time limits; free-facility responses (Paradigms A, C1-free, C2) will be re-solved to tighter gaps on high-performance infrastructure for the final version. (x) *Scope*. Wisconsin-specific magnitudes do not transfer verbatim to other Midwest states (feedstock

mix, demand, and dairy waste streams differ); the qualitative policy mechanisms are the transferable findings.

5 Conclusion

We presented a multi-product, market-clearing supply chain model for a Wisconsin biochar CDR system and embedded three carbon policy architectures in it. We report three findings. First, policy design dominates policy stringency for negative-emission systems: a linear carbon tax and a net emission cap both fail to move the optimum of a demand-constrained, net-negative network, while throughput-scaled removal credits activate the bulk biochar market at \$25/tCO₂e and mobilize essentially all near-term biomass at \$200/tCO₂e, increasing system surplus from \$0.71 B to \$3.36 B/yr and deepening net removal from 1.60 to 1.96 Mt CO₂e/yr. Second, the endogenous marginal abatement cost traced by the cap-and-trade sweep ($\approx$ \$351–391/t CO₂e over the first binding intervals, rising to $\approx$ \$599/t) lies above EU ETS allowance prices and the voluntary removal-credit band, quantifying how expensive gross-emission abatement is for a pyrolysis-based system, and why removal crediting, not gross-emission pricing, is the effective policy lever. Third, regulatory detail measurably reshapes technology: the VM0044 $H/C \leq 0.7$ gate excludes all torrefaction-like 300°C biochar from crediting and converts the system to 500°C pyrolysis as credit prices rise (2.25 Mt near-term, 4.25 Mt mature-market at \$200/tCO₂e), and quality-differentiated demand alone sustains 0.8 Mt/yr of 500°C biochar in the no-policy baseline.

For Wisconsin and the Midwest, the practical message is that feedstock is not the constraint; demand is. Policies that underwrite farmer adoption or monetize durable carbon at voluntary-market rates expand the system far more effectively than supply-side measures, and the spatial structure of value (biochar inherent values of \$275–324/t across scenarios) implies that location rents are sensitive to how much biomass the market ultimately calls for. Extensions under development include seasonal inventory and multi-period investment, the integration of anaerobic-digestion waste streams that characterize the dairy economy, and tighter optimality certification on high-performance computing for the free-facility policy sweeps.

A Data sources and processing

A.1 Biomass availability

County-level feedstock availability (dry tonnes/yr) comes from the DOE 2023 Billion-Ton Report (BT23) county-level downloads (bioenergykdf.ornl.gov), agricultural and forestry modules; the BT23 forestry module draws on USFS Forest Inventory and Analysis (FIA) data, which is therefore the underlying forestry source cited in the main text. The baseline uses the *near-term* scenario; the *mature-market medium* scenario serves as sensitivity. The BT23 corn stover series is constructed from grain-corn production via a 1:1 stover-to-grain ratio and is therefore *exclusive of silage corn acreage by construction* (silage is absent from the POLYSYS crop list; the 2022 Census of Agriculture reports 0.81 M acres of corn silage versus 3.23 M acres of grain corn harvested (20.1% of corn acreage), so

erroneously including silage would inflate stover by roughly 25%; the county-level grain/silage split is archived for a silage-inclusive sensitivity). The download field `price_offered` is the farmgate bid threshold (2022 USD per dry short ton, ash-inclusive, excluding transport); the near-term scenario represents approximately 2030 conditions with residues only and a \$70/dry-ton price ceiling. The tonnages are theoretical availability: field-level collection losses, residue-retention requirements, and landowner participation are not modeled (a -40% availability sensitivity is reported in the main text). Feedstock mapping: corn stover, miscanthus, switchgrass, willow, and poplar map directly from the agricultural download; logging residues aggregate the five BT23 logging-residue classes (hardwood lowland/upland, mixedwood, softwood natural/planted); small-diameter trees aggregate the four SDT classes; mill residues aggregate hardwood and softwood processing residues; other forest waste corresponds to “forest waste human generated.” Wheat, oats, barley, and soybean straw are taken from USDA NASS 2022 Census of Agriculture county production data converted to theoretical residue tonnage (NASS Census residue estimates, archived in the project workbook sheet `02_NASS_Validation`; Table S1 gives the county corn grain/silage acreage split), because BT23 reports near-zero availability for these resources at \$70/t. State totals: near-term 6.97 Mt dry/yr; mature-market medium 13.53 Mt dry/yr (of which the nine modeled BT23 categories account for 12.54 Mt). BT23 *waste* resources are characterized but not yet modeled: the full Wisconsin wastes download (16 categories, 72 counties, five scenarios; archived as `bt23_wastes_WI_summary.csv`) gives 2.27 Mt dry/yr at $\leq \$70/\text{t}$ in the near-term scenario, dominated by dairy manure (1.13 Mt), clean urban wood (0.34 Mt), textiles (0.27 Mt), and yard waste (0.25 Mt), with sludge, paper, plastics, and FOG at negligible $\leq \$70/\text{t}$ availability; these form a co-location extension for Wisconsin’s dairy and paper economy (Section Limitations) and are an explicit model-boundary statement, not an oversight of data.

Table S1: USDA NASS 2022 Census of Agriculture, Wisconsin corn acreage split (archived `corn_grain_silage_county_2022.csv`). The BT23 grain-corn stover series excludes silage by construction; a silage-inclusive stover estimate would add roughly 25%.

	acres (M)	share of corn acreage
Grain corn harvested	3.23	79.9%
Corn silage	0.81	20.1%

A.2 Feedstock characterization

Table S2 gives, per feedstock: initial moisture content (12–45%), drying yield to 10% moisture $\eta = (100 - \text{MC})/90$, farmgate prices (\$40–60/dry t, wet-adjusted), 300°C biochar yield (0.33–0.38 t/t), carbon content (0.41–0.50), and molar H/C (1.00–1.40 at 300°C, 0.40–0.52 at 500°C). Moisture contents follow the feedstock-collection literature compiled in Roberts et al. [17] (corn stover 15–30%, woody biomass 45%) and the forestry-drying review of Fagernas et al. [9]; char carbon contents are cross-checked against the feedstock-class means of Woolf et al. [28] (SI Table S4: cereals 59.9%, herbaceous crops 56.8%, wood 75.0%, green waste 67.5%). Yields are expressed per tonne of 10%-moisture dried

feedstock (the model’s dry product); if oven-dry-basis literature yields are preferred, all yield and per-dry GHG factors scale by 0.9.

Table S2: Feedstock characterization: moisture (MC), dry carbon content C_{dry} , biochar mass yields y_{300}/y_{500} , molar H/C at 300/500°C, and farmgate price. H/C values follow the literature audit of Section A.3 (300°C: Rafiq et al. 2016 and the Ippolito et al. 2020 meta-analysis band; 500°C: Rafiq 2016, Enders 2012, Ippolito 2020): all 300°C chars exceed the VM0044 $\text{H/C} \leq 0.7$ gate (torrefaction-like, never credit-eligible), and all 500°C chars satisfy it.

Feedstock	class	MC (%)	C_{dry}	y_{300}	y_{500}	H/C ₃₀₀	H/C ₅₀₀	farmgate (\$/dry t)	Reference
Corn stover	ag	17.5	0.45	0.38	0.300	1.40	0.49	60	[16, 11]
Soybean straw	ag	14.0	0.43	0.36	0.284	1.30	0.50	60	[11]
Wheat straw	ag	12.5	0.42	0.35	0.276	1.25	0.51	60	[11]
Oats straw	ag	13.0	0.41	0.34	0.268	1.25	0.52	60	[11]
Barley straw	ag	12.0	0.41	0.34	0.268	1.25	0.51	60	[11]
Switchgrass	ag	12.5	0.47	0.38	0.300	1.30	0.47	60	[11]
Miscanthus	ag	15.0	0.48	0.37	0.292	1.30	0.45	60	[11]
Poplar	for	45.0	0.50	0.35	0.276	1.10	0.40	60	[28, 8]
Willow	for	45.0	0.50	0.35	0.276	1.10	0.40	60	[28, 8]
Logging residues	for	42.5	0.49	0.33	0.261	1.05	0.42	40	[9, 8]
Small-diameter trees	for	45.0	0.49	0.33	0.261	1.05	0.41	60	[17, 8]
Mill residues	for	35.0	0.50	0.34	0.268	1.00	0.40	50	[8]
Other forest waste	for	20.0	0.47	0.35	0.276	1.15	0.45	50	[11]

A.3 Technology economics

T1 dehydration: OPEX \$5.2–10.8/wet t (moisture-dependent), CAPEX \$0.8/2.5/7.0 M for 33/165/660 kt/yr plants [30, 10]. T2 slow pyrolysis: OPEX \$30/dry t, CAPEX \$9/21.5/51 M for 60/198/660 kt/yr [20, 17]. The 300°C/500°C yield and carbon-content pairs for corn stover (0.38/0.30 t/t; 0.45/0.55) are the advisor-confirmed technology-table values (V9 workbook, sheet 02); per-feedstock 500°C values use the documented scaling $y_{500} = 0.789 y_{300}$, $C_{500} = 1.222 C_{300}$. Source audit: Roberts et al. [17] is a 450°C slow-pyrolysis LCA and contains no 300°C or 500°C case; it supports $y_{500} = 0.30$ indirectly, via its char-yield range of 28.8–33.0 wt% (citing Antal and Grønli 2003, *Ind. Eng. Chem. Res.* 42:1619), it matches the Rafiq et al. corn-stover 500°C yield of 29.2% [16], and it supports $\rho_{500} = 0.80$ through its 80% stable-carbon assumption. Woolf et al. [28] do not tabulate temperature-series yields or H/C ratios. The H/C values are taken from the temperature-series literature (Table S2): 300°C chars have $\text{H/C} = 1.00$ –1.40 (torrefaction-like; Rafiq et al. [16] measure 1.41 for corn stover at 300°C, and the Ippolito et al. meta-analysis [11] places sub-400°C chars in the “partially thermochemically altered” range), so no 300°C biochar meets the VM0044 $\text{H/C} \leq 0.7$ gate, and its credit rate is zero throughout. The 500°C chars have $\text{H/C} = 0.40$ –0.52 (Rafiq et al. [16]: corn stover 0.49, benchmarked against Enders et al. [8]; the Ippolito et al. meta-analysis 500–599°C band mean is ≈ 0.54), comfortably below the gate, so all 500°C biochar is credit-eligible. A ± 0.1 H/C

perturbation changes no eligibility outcome (Section A.7); the first flip of a 500°C char requires +0.25, which is the stress test reported there. The $C_{500} = 0.55$ value is conservative relative to the published 500°C carbon contents of 62.5% (Ippolito et al. [11] band mean) and 64.5% (Rafiq et al. [16]), so 500°C credit rates are conservative lower bounds.

A.4 Greenhouse gas factors

Transport 0.0902 kg CO₂e/t-km (32-t truck) [27]; collection and baling 35 kg/t dry [30]; drying 25 kg/t H₂O [10, 9]; slow pyrolysis 150 kg/t dry net of a syngas electricity credit at 50–65 kWh/t and 0.57 kg/kWh Wisconsin grid intensity (Roberts et al. [17]); (the pyrolysis term is the dominant process emission and carries the largest parameter uncertainty). Carbon removal is defined as a positive quantity, $r_t^{\text{seq}} = \rho_t C_t (44/12) \times 10^3$ kg CO₂e/t biochar, with permanence $\rho_{300} = 0.65$, $\rho_{500} = 0.80$ per VM0044 v1.2 Table 3 (IPCC 2019 Table 4AP.2: 0.65 for 350–450°C, 0.80 for 450–600°C, 0.89 for >600°C; the 300°C value uses the lowest tabulated band, 350–450°C, and the permanence of sub-350°C chars may be lower, so ρ_{300} should be read as an upper-bound parameter); N₂O suppression 52 kg/t [4]; decomposition baseline $b_f = 0.9 C_f^{\text{dry}} (44/12) \times 10^3$ kg/t dry. Carbon credits are intangible certificates: they carry zero transport cost and are *not* assigned transport emissions (the certificates are settled at the producing county).

A.5 Demand data pack

The three-segment biochar demand curve is calibrated to: (i) the U.S. wholesale price band \$600–2,778/t (mode $\approx$ \$1,000/t); (ii) production-side costs \$150–400/t; (iii) Wisconsin farmland of 14.2 M acres with application rates of 5–20 t/ha and adoption below 100%, so that the 4.16 Mt aggregate of the previous model version is treated as an optimistic peak; and (iv) a non-soil sink valued at \$30/t (construction/filtration media) with a landfill disposal sensitivity of –\$45 to –\$84/t. Segment capacities are allocated across counties in proportion to farmland area; the CDR-premium segment is interpreted as access to national corporate buyers through credit markets rather than local Wisconsin demand, and the farmland proportionality is a spatial proxy (county-specific adoption data are not available). The 0.8 Mt premium capacity is a national-market allocation: current U.S. biochar purchases reported by CDR.fyi are in the hundreds of thousands of tonnes per year [5], so the segment assumes the several-fold market growth projected for biochar credits rather than today’s volumes. The decomposition fraction 0.9 is an upper bound (immediate, complete decay); a sensitivity to 0.7/0.5 scales the Paradigm A credit rates proportionally and is flagged in the Limitations section.

A.6 Solver settings

Gurobi 13.0.2 (academic license), 8 threads. Step 1 MIP: 1% gap, 1,800 s, MIPFocus=1, NoRelHeurTime=60. Policy sweeps: 0.5% gap target, 600 s warm-started from the previous layout (Paradigm A credit prices $p_c = 25/50$ re-solved with 1,800 s limits, improving the incumbents); sweep points that hit the time limit are feasible incumbents (reported gaps 0.5–4.7%, Table S3). All duals read from fixed-layout LPs solved with *dual simplex* (exact vertex shadow prices). The cap’s vertex dual is reported

with Gurobi’s sign convention for $\leq$ constraints in maximization (the negative of the marginal value): $-\$374/\text{t}$ at the first binding cap (near-term), inside the $\$351\text{--}391/\text{t}$ finite-difference band of the first binding intervals; the endogenous carbon price is reported as both. The GHG LP (Step 3) pins demand quantities to the Step-2 equilibrium (file `dem_star_v2.csv`). Paradigm C1 is solved both with fixed facilities (LP) and with free facilities (MIP, C1-free), and Paradigm A and C2 use free-facility MIPs throughout. The mature-market $p_c = \$100$ point of the Paradigm A sweep was additionally re-solved for 1,800 s at a 0.1% gap target (`policy_A_recert_v2.csv`); its 500°C share improved from 1.75 to 2.79 Mt at a 1.9% gap.

Table S3: Optimality-gap report: free-facility MIPs solve at a 0.5% gap target (Step 1 at 1%); points that hit the time limit are feasible incumbents, so their reported surplus is a lower bound on the true optimum. v2b runs record termination status only (†), without gap percentages.

experiment (time limit)	near-term	mature-market medium
Step 1 MIP (1,800 s, 1% target)	OPTIMAL	OPTIMAL
Paradigm A sweep (600 s; $p_c = 25/50$ at 1,800 s)	0.95–1.9%	0.5–3.2%
C1-free (600 s)	1.1–1.2%	OPTIMAL
C2 sweep (600 s; extended points at 1,800 s)	1.6–4.7%	OPTIMAL–2.5%
<i>Quality-differentiated model (v2b)</i>		
Step 1 MIP (1,800 s, 1% target)	TIME_LIMIT †	OPTIMAL
Paradigm A sweep (600 s)	TIME_LIMIT †	OPTIMAL–TIME_LIMIT
C2 sweep (600 s)	TIME_LIMIT †	OPTIMAL–TIME_LIMIT

A.7 Eligibility sensitivity: H/C uncertainty and the gate

The VM0044 gate is applied to the literature-consistent H/C values of Table S2. A ± 0.1 perturbation around those central values changes *no* eligibility outcome: every 300°C value remains above the gate (1.00–1.40 minus 0.1 is still $\geq 0.90 > 0.7$), and every 500°C value remains below it (0.40–0.52 plus 0.1 is still $\leq 0.62 < 0.7$). The first flip of a 500°C char therefore requires a +0.25 stress, well beyond the published 500°C spread of 0.40–0.55, which pushes the agricultural-residue chars (corn stover $0.49 \rightarrow 0.74$; soybean/wheat/oats/barley/switchgrass ≥ 0.72) over the gate while forestry and miscanthus chars ($0.40\text{--}0.45 \rightarrow 0.65\text{--}0.70$) remain eligible. Table S4 re-solves Paradigm C2 under that stress. At $p_c = \$50$ nothing changes in either scenario. At $p_c = \$200$ the canonical full conversion is curtailed: near-term production falls back to 2.30 Mt of which only 0.31 Mt is 500°C (credits 0.55 Mt, surplus $\$716\text{ M}$ versus $\$1,141\text{ M}$ canonically), and mature-market production falls back to 2.30 Mt of which 1.59 Mt is 500°C (surplus $\$1,018\text{ M}$ versus $\$1,501\text{ M}$). Two conclusions follow. First, the 300°C exclusion is over-determined: the nearest measured value is 0.9, two tenths above the gate, so no plausible H/C measurement error re-qualifies torrefaction-like 300°C chars. Second, the C2 technology-conversion result is robust within the published 500°C H/C band but degrades if agricultural-residue 500°C chars were, contrary to every measurement we found, above 0.7, which is why VM0044 and IBI treat the gate as a per-batch measured property rather than a design assumption.

Table S4: Paradigm C2 under the +0.25 H/C stress (agricultural-residue 500°C chars ineligible): surplus, biochar, and 500°C share at $p_c = \$50$ and $\$200/\text{tCO}_2\text{e}$, versus the canonical values (Table S2 H/C).

		surplus (M\$)	BC (Mt)	BC500 (Mt)	CC (Mt)
near-term	canonical, \$50	653.3	2.30	0.00	0.00
	stress, \$50	653.4	2.30	0.00	0.00
	canonical, \$200	1,141.1	2.25	2.25	3.64
	stress, \$200	715.9	2.30	0.31	0.55
mature-market	canonical, \$50	684.6	2.30	0.50	0.86
	stress, \$50	684.5	2.30	0.50	0.86
	canonical, \$200	1,501.0	4.25	4.25	7.03
	stress, \$200	1,018.1	2.30	1.59	2.76

A.8 Reproducibility

One-at-a-time sensitivity of the baseline surplus is reported in S-Figure S1: demand-price shocks dominate ($\pm 39\%$ of surplus for $\pm 20\%$ of price), farmgate and transport shocks are minor, and a -40% feedstock-availability shock (a proxy for field-collection losses) costs 29% of near-term surplus but only 4% in the mature-market scenario.

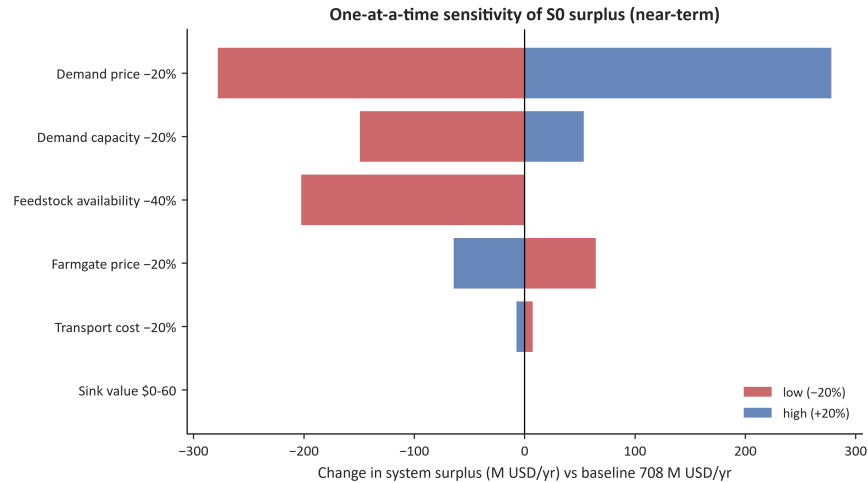


Figure S1: One-at-a-time sensitivity of the baseline surplus to $\pm 20\%$ parameter changes (LP with the baseline facility layout, near-term scenario). Demand-side levers (price, capacity) dominate the response, farmgate and transport parameters are minor, and the -40% availability shock is the only supply-side perturbation of comparable size; the system is demand-constrained.

Data generation (`generate_data_v2.py`, `generate_data_v2b.py`), model (`01-06_*.jl`), and figures (`visualize_v2.py`, `fig0/1/45/9_*.py`) scripts are archived with the manuscript; the code and data will be released in a public repository upon publication and are available from the corresponding author during review. All optimization models are implemented in Julia 1.12 with JuMP and solved with Gurobi 13.0.2 (academic license) on an 8-thread machine; solve times are 1,800 s

for the Step-1 MIP, 600–1,800 s per policy sweep point, and seconds for the fixed-layout LPs. Figure scripts use matplotlib with the Calibri family at 400 DPI; county boundaries and the Great Lakes layer are drawn from U.S. Census Bureau and Natural Earth shapefiles.

A.9 Proof of Proposition 1

For the fixed-layout LP, the cap constraint $E^+(x) \leq \text{CAP}$ is appended to (2.1) with the layout variables z fixed. Strong duality (Assumption 1) and the differentiability of the LP value function give, by the standard parametric-LP sensitivity theorem, $\mu_{\text{CAP}} = \partial \Pi^* / \partial \text{CAP}$, where μ_{CAP} is the optimal dual variable of the cap constraint. Gurobi reports the dual of a $\leq$ constraint in a maximization with the opposite sign convention (it reports the reduction in objective per unit of constraint tightening), i.e., the reported value is $-\partial \Pi^* / \partial \text{CAP}$. The endogenous carbon price (marginal abatement cost) is therefore the negative of the reported dual, and the finite-difference estimates $\Delta \Pi^* / \Delta \text{CAP}$ in Table S3 reproduce it to numerical tolerance (-374 vs. $\$351\text{--}391$ $\$/\text{t}$ over the first binding intervals, near-term).

A.10 Proof of Proposition 2 (two-regime crossover)

Fix feedstock f and a biochar market price P , and consider the two fixed-yield routes $r \in \{3, 5\}$ with dry-biomass yields γ_f^r , credit rates ρ_f^r (tCC/t dry), and chain cost ω independent of the route (farmgate is per dry tonne; the route-specific OPEX is already absorbed in ω ; results are unchanged if route-specific costs $\omega_3 \neq \omega_5$ are used, with $\omega(1/\gamma^5 - 1/\gamma^3)$ replaced by $\omega_5/\gamma^5 - \omega_3/\gamma^3$).

Demand-bound. When feedstock is abundant, the active constraint is the demand bid: the model serves a fixed biochar quantity Q , and the supply path with the higher surplus per tonne of biochar served wins. Per tonne served, the route- r chain consumes $1/\gamma_f^r$ dry tonnes, paying ω/γ_f^r and earning credit revenue $(\rho_f^r/\gamma_f^r) p_c$, so

$$\Pi^r/Q = P - \omega/\gamma_f^r + \rho_f^r p_c/\gamma_f^r. \quad (\text{A.14})$$

Subtracting the $r = 3$ expression gives $\Delta_f^{\text{dem}}(p_c)$ as in the statement, which is affine in p_c with slope $\rho_f^5/\gamma_f^5 - \rho_f^3/\gamma_f^3 > 0$ (the 500°C credit rate per served tonne exceeds the 300°C one; under the VM0044 gate $\rho_f^3 = 0$ this is automatic). It crosses zero at $\bar{p}_f^{\text{dem}}$, which is independent of P — P cancels in the difference—and positive whenever the slope is positive. The crossing is unique because the difference is affine.

Supply-bound. When feedstock is scarce, the active constraint is the feedstock availability: a fixed dry mass M is processed, and the route with the higher surplus per dry tonne wins. Per dry tonne, route r yields γ_f^r t biochar at price P and earns credit revenue $\rho_f^r p_c$, so

$$\Pi^r/M = \gamma_f^r P + \rho_f^r p_c - \omega. \quad (\text{A.15})$$

Subtracting gives $\Delta_f^{\text{sup}}(p_c) = (\gamma_f^3 - \gamma_f^5)P - (\rho_f^5 - \rho_f^3)p_c$, affine in p_c with *negative* slope, crossing zero at $\bar{p}_f^{\text{sup}}$, strictly increasing in P (the yield gap $\gamma_f^3 - \gamma_f^5 > 0$ multiplies P), and unique. Note the sign asymmetry: in the demand-bound regime the credit slope is positive (500°C wins at high p_c because its lower yield per dry tonne concentrates credit revenue per served tonne), while in the supply-bound

regime the credit slope is negative (the yield loss is a direct revenue loss at price P , and credits must compensate it per dry tonne).

Numerical values. Using the pooled feedstock averages of the LCFS-style baseline-and-credit rates adopted in Paradigm A ($\bar{\gamma}^3 = 0.352$, $\bar{\gamma}^5 = 0.278$ t/t dry; $\bar{\rho}^3 = 1.771$, $\bar{\rho}^5 = 1.849$ tCC/t dry, i.e. a credit-rate premium of $\bar{\rho}^5 - \bar{\rho}^3 \approx 0.077$ tCC/t dry; combined chain cost $\omega \approx \$130$ /t dry) at the bulk-segment price $P = \$250$ /t:

$$\bar{p}^{\text{dem}} = \frac{130 (1/0.278 - 1/0.352)}{1.849/0.278 - 1.771/0.352} \approx \$61/\text{t}, \quad \bar{p}^{\text{sup}} = \frac{(0.352 - 0.278) 250}{1.849 - 1.771} \approx \$239/\text{t}. \quad (\text{A.16})$$

Both values are confirmed by the paradigm-A sweep: the mature-market (demand-bound) case switches to 500°C at $p_c \approx \$61/\text{t}$ and the near-term (supply-bound) case stays on 300°C through $p_c = \$200/\text{t}$ with crossover indicated near $p_c \approx \$239/\text{t}$ (Fig. 10, which reports the same closed-form crossings with the feedstock averages). $\square$

A.11 Proof of Proposition 3 (tax invariance in the demand-constrained regime)

Write the fixed-layout LP in the compact form

$$\Pi^* = \max_{x \in X} c^\top x, \quad X = \{x : Ax \leq b, x \geq 0\}, \quad (\text{A.17})$$

where the demand stack rows are $d \mapsto \text{dem}_d \leq \bar{d}_d$ with objective coefficient λ_d^{dem} in c , and the mass-balance rows carry duals π . Suppose x^* is an optimal basic solution with the property that the active demand rows are exactly those with $\pi_n^{\text{BC}} < \lambda_d^{\text{dem}}$ (strict slack at the margin) in the terminology of the statement—the demand-constrained regime. Consider the perturbed problem

$$\Pi_\tau^* = \max_{x \in X} c^\top x - \tau e^\top x, \quad e \geq 0 \text{ (gross-emission rates per activity unit)}. \quad (\text{A.18})$$

The perturbed objective is $(c - \tau e)^\top x$. We show $\arg \max$ is unchanged. Write $x^* = \sum_{\text{basic } j} w_j x_j$; since x^* is an LP vertex with basis B , $x_B^* = B^{-1}b$ is determined by the active constraints alone, so any perturbation of the objective—which does not change the constraint matrix A —cannot change the vertex x_B^* ; it can only change *whether* that vertex is optimal. Optimality persists iff the reduced costs of the nonbasic activities j remain nonpositive for the current basis: $c_j^{\text{red}} - \tau e_j^{\text{red}} \leq 0$ for all nonbasic j . For the binding-demand rows, the reduced cost of the corresponding supply path q entering the basis is exactly $c_q^{\text{red}} = \lambda_{d(q)}^{\text{dem}} - \pi_{n(q)}^{\text{BC}} > 0$ (the price gap at the margin); the path is already at its capacity bound and entering more of it is blocked. All other nonbasic activities have $c_j^{\text{red}} \leq 0$ (they are unprofitable at the current duals), so they remain unattractive as long as τe_j^{red} does not flip the sign of a positive reduced cost. Because the only positive reduced costs are the marginal supply paths with $\lambda^{\text{dem}} - \pi^{\text{BC}} > 0$, the critical tax rate is

$$\tau_{\max} = \min_{q \text{ marginal}} \frac{\lambda_{d(q)}^{\text{dem}} - \pi_{n(q)}^{\text{BC}}}{e_q^+} \quad (\text{A.19})$$

as stated; for $\tau < \tau_{\max}$ the optimal basis, and hence the material flows, technology mix, and cleared quantities, are identical to x^* , while the objective decreases by $\tau E^+(x^*)$ —the tax is a pure surplus-to-government transfer. Numerically, the marginal price gap is $\lambda^{\text{M}} - \max_n \pi_n^{\text{BC}} = 500 - 344.85 \approx \$155/\text{t}$

biochar and the gross-emission intensity of the marginal supply path is ≈ 0.44 t CO₂e/t biochar ($E^+ = 1,002.8$ kt over 2.30 Mt served; T1 31–35 kg/t wet, T2 ≈ 121 kg/t dry, plus transport), giving $\tau_{\max} \approx \$350/\text{t CO}_2\text{e}$, above the top tested rate of $\$200/\text{t}$, consistent with the null dispatch response of C1 and C1-free (Table 2). The tiered slope does not matter: a convex piecewise-linear tax with positive slopes is, on the active paths, locally linear, so the argument applies tier by tier. $\square$

A.12 Nomenclature

Table S5: Nomenclature: sets, parameters, and variables of the base network model (2.1). Units: \$ = 2022 USD, t = metric tonne.

Type	Notation	Description
Sets	$\mathcal{N}$	County nodes ($ \mathcal{N} = 72$)
	$\mathcal{F}$	Feedstocks ($ \mathcal{F} = 13$: 7 agricultural, 6 forestry)
	$\mathcal{T}$	Technologies (T1 dehydration; T2 slow pyrolysis at 300/500°C; 39 total)
	$\mathcal{P}$	Products (wet biomass, dry biomass, biochar, carbon credits)
	$\mathcal{S}, \mathcal{D}$	Supply and demand bid rows (farmgate offers, demand segments)
	$\mathcal{A}$	Transport arcs (radius-limited)
	$\mathcal{K}$	Facility scale tiers ($k \in \{1, 2, 3\}$)
	$\text{ref}(t)$	Reference (input) product of technology t
Parameters	$\mathcal{S}_n, \mathcal{D}_n$	Supply/demand rows located at node n
	$\lambda_d^{\text{dem}}, \lambda_s^{\text{sup}}$	Demand bid (\$/t) and farmgate offer (\$/t)
	ω_t	Technology OPEX (\$/t of reference product)
	$\alpha_p^{\text{var}}, \alpha_p^{\text{fix}}$	Variable (\$/t-km) and fixed (\$/t) transport cost of product p
	$\kappa_{t,k}$	Annualized CAPEX of technology t at scale k (\$)
	$\bar{q}_{t,k}$	Capacity of technology t at scale k (t/yr)
	$\gamma_{t,p}$	Fixed-yield transformation coefficient of product p by technology t
	$\bar{d}_d, \bar{s}_s$	Demand and supply capacities (t/yr)
Variables	AF	Annualization factor (0.1175; 10%, 20 yr)
	$z_{ntk} \in \{0, \dots, 5\}$	Number of facilities of technology t , scale k , at node n
	$x_{n,p,t}$	Net production of product p by technology t at node n (t/yr)
	f_{ijp}	Transport flow of product p from i to j (t/yr)
	$\text{dem}_d, \text{sup}_s$	Cleared demand and supply (t/yr)
	d_{np}, s_{np}	Aggregated demand/supply at node n for product p
	p_{np}	Aggregated production at node n for product p
	π_{np}	Dual of the mass balance = inherent value of product p at node n (\$/t)

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